

A Vision-Language Foundation Model for Precise and Comprehensive Brain Tumor Diagnosis from Preoperative Multimodal Data

Yinong Wang^{1, †}, Jianwen Chen^{1, †}, Zhou Chen^{2, †}, Shuwen Kuang^{3, †}, Haoning Jiang¹, Yanzhao Shi¹, Huichun Yuan⁴, Yan-ran (Joyce) Wang⁵, Bing Wang⁷, Lei Wu⁸, Bin Tang⁹, Li Meng¹⁰, Baihua Luo¹¹, Bin Zhou^{6,12}, Wei Ding¹³, Weiming Zhong¹⁴, Wei Hou¹⁵, Yuanbing Chen¹⁶, Zhiping Wan¹⁷, Wei Wang¹⁸, Zhenkun Xiao⁷, Wenwu Wan¹⁹, Allen He²⁰, Yuyin Zhou²¹, Longbo Zhang^{2,6,24}, Feifei Wang²², Zhixiong Liu^{2,6,24}, Michael Iv²³, Xuan Gong^{2,6,24, *}, and Liangqiong Qu^{1, *}

¹School of Computing and Data Science, The University of Hong Kong, China

²Department of Neurosurgery, Xiangya Hospital, Central South University, China

³Department of Oncology, Xiangya Hospital, Central South University, China

⁴Changde Hospital, Xiangya School of Medicine, Central South University (The First People's Hospital of Changde City), China

⁵Department of Biomedical Data Science, School of Medicine, Stanford University, Stanford, USA

⁶Department of Neurosurgery, Xiangya Hospital, Central South University, Jiangxi (National Regional Center for Neurological Diseases), China

⁷Department of Neurosurgery, Hengyang Medical School, University of South China, China

⁸Department of Neurosurgery, The Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, China

⁹Department of Neurosurgery, The First Affiliated Hospital, Jiangxi Medical College, Nanchang University, China

¹⁰Department of Radiology, Xiangya Hospital, Central South University, China

¹¹Department of Pathology, Xiangya Hospital, Central South University, China

¹²Department of Neurosurgery, Jiangxi Provincial People's Hospital, The First Affiliated Hospital of Nanchang Medical College, China

¹³The Affiliated Children's Hospital of Xiangya School of Medicine, Hunan Children's Hospital, China

¹⁴Department of Neurosurgery, Shenzhen Second People's Hospital, China

¹⁵Department of Neurosurgery, First Hospital of Lanzhou University, China

¹⁶Department of Neurosurgery, The Third Xiangya Hospital, Central South University, China

¹⁷Department of Neurosurgery, Tongji Hospital, School of Medicine, Tongji University, China

¹⁸Department of Radiology, Tongji Hospital, School of Medicine, Tongji University, China

¹⁹Department of Neurosurgery, Chongqing Traditional Chinese Medicine Hospital, China

²⁰Basis International School Park Lane Harbour, China

²¹Department of Computer Science and Engineering, University of California, Santa Cruz, USA

²²Department of Electrical and Electronic Engineering, The University of Hong Kong, China

²³Department of Radiology, School of Medicine, Stanford University, Stanford, USA

²⁴National Clinical Research Center of Geriatric Disorders, Xiangya Hospital, Central South University, China

*Corresponding authors. Email addresses: liangqqu@hku.hk (Liangqiong Qu), gong.xuan@csu.edu.cn (Xuan Gong)

[†]These authors contributed equally to this work.

ABSTRACT

Manual interpretation of multi-parametric Magnetic Resonance Imaging (MRI) for precise brain tumor diagnosis can be challenging, labor-intensive, and prone to inter-observer variability. Here, we present BrainVLM, a vision-language foundation model that delivers precise and reliable brain tumor classification with diagnostic uncertainty quantification and radiology reports generation. We developed and validated BrainVLM using 46,502 patients collected from 36 online repositories and 12 medical hospitals, covering all 12 World Health Organization (WHO) classified brain tumor types and most subtypes. In primary evaluations, BrainVLM achieved an area under the curve (AUC) of 0.91, and an F_1 score of 0.83, surpassing neuroradiologists ($F_1 = 0.81$). External validation across 11 independent medical centers demonstrated strong generalizability (AUC = 0.87, $F_1 = 0.76$ vs. neuroradiologists' 0.71). Notably, BrainVLM's uncertainty quantification mechanism further identifies 70% of the incorrect predictions through low-confidence predictions (probability < 0.85). The supporting radiology reports significantly outperformed existing models in both linguistic coherence and clinical relevance (RaTEScore 0.73 vs. 0.61). In a blinded multireader study on a 123-case subset, BrainVLM-augmented neuroradiologist evaluations demonstrated a statistically significant 23.3% improvement in F_1 score compared to unaided assessments ($P < 0.001$), while reducing diagnostic time by 36.1%. In prospective real-world evaluation, BrainVLM demonstrated a statistically significant 4.7% higher F_1 score compared with neuroradiologists. For molecular subgroup prediction in adult-type diffuse gliomas, BrainVLM achieved an AUC of 0.95 and 0.93 in primary and external validation, respectively, highlighting its strong potential for downstream molecular classification tasks. Together, by integrating

diagnostic precision with radiology reports and uncertainty quantification, BrainVLM bridges the gap between AI automation and clinical trust, offering a scalable solution to enhance neuro-oncology workflows.

Introduction

Brain neoplasms can have long-lasting and life-altering physical, cognitive, and psychological impacts on a patient's life, occurring in all anatomical regions of the encephalon and adjacent structures. Malignant brain tumors represent one of the most common cancers and the second leading cause of cancer-related death among children and adolescents, making them a prevalent healthcare burden¹⁻³. The 2021 fifth edition of the World Health Organization Classification of Central Nervous Tumors (WHO CNS5) identifies 12 major brain tumor types and over 100 distinct tumor subtypes, ranging from benign and indolent neoplasms such as grade 1 meningioma to malignant and aggressive tumors such as grade 4 glioblastoma⁴. Common treatment options for brain tumors include neurosurgical resection, radiation therapy, and chemotherapy, with surgery serving as the primary initial management in most cases^{5,6}. However, the choice of treatment modality, surgical approach, and the extent of resection can depend on the initial presurgical diagnosis⁷. For example, maximal safe gross total resection is generally recommended for patients with diffuse gliomas⁷. In contrast, lymphomas and some germ cell tumors may be managed effectively with chemotherapy and/or radiation therapy without surgical resection^{8,9}. Hence, a prompt and accurate initial diagnosis of brain tumors is an important first step to ensure reliable and optimal treatment for brain tumors.

Magnetic resonance imaging (MRI) is the primary diagnostic modality for presurgical diagnosis and longitudinal monitoring of patients with brain tumors¹⁰. Current diagnostic practice requires neuroradiologists to review multi-parametric MRI scans to identify and diagnose potential lesions and convey their findings and impression in a radiology report. Based on these MRI findings, clinicians subsequently determine the optimal therapeutic pathway, formulate comprehensive surgical planning strategies, and evaluate the surgical risk and prognosis. However, this manual interpretation is time-consuming and error-prone due to overlapping radiologic characteristics of different brain tumors, inter-observer variability, and potential perceptual errors. This problem is further compounded by the global shortage of specialist neuroradiologists due to the extensive training efforts required to gain expertise¹¹. These unmet needs call for advanced techniques to develop an automatic brain tumor diagnostic tool for rapid and precise diagnoses and report generation, thereby accelerating the time-sensitive diagnosis process and reducing clinician workload.

Artificial intelligence (AI), particularly the recently emerged vision language models (VLMs)¹²⁻¹⁵, has the potential to advance automatic disease diagnosis¹⁶⁻¹⁸. Compared to the flourishing development of AI diagnostic frameworks for other tumors, comprehensive AI models for presurgical brain tumor diagnosis remain significantly underexplored^{19,20}. At least two critical challenges hinder the widespread clinical adoption of AI-based brain tumor diagnosis¹⁷. First, current publicly available datasets for brain tumor analysis—exemplified by the BraTS consortium datasets—suffer from a narrow focus, predominantly featuring gliomas, meningiomas, and metastases, while underrepresenting other tumors in the WHO CNS5²¹⁻²³. This limited diversity is further compounded by their reliance on pure imaging data, which lacks radiology reports necessary for holistic clinical interpretation. Second, existing AI models predict brain tumor types without assessing the confidence and reliability of the prediction or explaining the underlying clinical rationale²⁴⁻²⁷. This issue is particularly pronounced in large language models (LLMs) or VLMs, which often exhibit unwarranted high confidence in erroneous predictions^{24-26,28}. This overconfidence in AI predictions, coupled with the opaque decision-making process, undermines clinicians' trust and poses a substantial barrier to their clinical translation and adoption.

In this study, we developed BrainVLM, the first AI model capable of comprehensive classification across all 12 WHO CNS5-defined brain tumor types and three sub-categories of gliomas, glioneuronal tumors, and neuronal tumors. Central to the BrainVLM framework are its clinician-trust focused innovations: 1) an integrated uncertainty quantification strategy explaining the degree of prediction reliability, and 2) a radiology report generation module elucidating the underlying clinical rationale. To support these achievements, we curated BrainTumor46K, the largest and most comprehensive multi-modal neuro-oncology dataset to date, aggregating 46,502 individuals from 36 public repositories and 12 medical centers. We evaluated BrainVLM's diagnostic capabilities by comparing its outputs with preoperative board-certified neuroradiologists' interpretations (derived from clinical radiology reports) and multiple state-of-the-art VLMs, using postoperative pathological diagnoses as the gold standard. We further conducted two proof-of-concept studies to validate its clinical utility in AI-clinician workflows: a blinded multi-reader study where 12 neuroradiologists across varying experience levels interpreted 123 retrospective cases with or without AI assistance, and a real-world prospective study, where 713 patients were simultaneously assessed by BrainVLM and radiologists before the experienced surgery. Our findings demonstrate that AI-learned diagnosis based on presurgical MRI scans can effectively assist clinicians' decision-making, showing the potential to substantially enhance early brain tumor diagnosis and significantly alleviate clinicians' workloads. We further evaluated the diagnostic performance of BrainVLM for adult diffuse gliomas (according to WHO CNS 5), the model achieved an AUC of 0.95 in 3-fold primary validation and an AUC of 0.93 in external validation (Fig. 3b). The results demonstrated that BrainVLM is robust for downstream brain tumor

105 diagnostic tasks. We will publicly release our training dataset to promote transparency, reproducibility, and future research in
106 the field of neuro-oncology AI.

107 Results

108 Datasets and study design

109 For the development of BrainVLM, we curated BrainTumor46K, a comprehensive multi-modal brain tumor dataset. It comprises
110 46,502 individuals aggregated from 36 public repositories and 12 collaborating medical centers (Fig. 1a, Table 1, Extended
111 Data Tables 1 and 2). The BrainTumor46K comprises 31,704 patients with pathologically confirmed brain tumors (20,929
112 from public repositories and 10,775 from medical centers) and 14,798 healthy controls (sourced from public repositories)
113 (Fig. 1a). Public repositories contributed two categories of brain tumor data: (1) 13,096 patients with 2D MRI slices paired
114 with text descriptions extracted from scientific publications or social media sources (e.g., figure captions, inline mentions,
115 or speech-to-text transcripts), and (2) 7,833 patients with 3D multi-parametric MRI scans, including T1-weighted (T1), T1
116 contrast-enhanced (T1c), T2-weighted (T2), and T2-FLAIR (FLAIR) sequences, linked to either expert-curated or AI-generated
117 radiology reports. The detailed public repositories and its pre-processing pipeline for the public repositories are summarized in
118 the methods section (see Extended Data Fig. 1). In addition to public repositories, data from collaborating medical centers
119 were collected, forming two institutional cohorts: (1) A multi-center retrospective cohort (10,062 patients, 48% male, 52%
120 female; mean (standard deviation, SD) age 46 (18) years) includes demographics, 3D multi-parametric MRI scans (T1, T1c,
121 T2, FLAIR), expert-curated radiology reports, and pathological diagnoses. (2) A prospective cohort sourced from Xiangya
122 hospital, including 713 patients (45% male, 55% female; mean (SD) age 45 (19) years). Fig. 1a and Extended Data Table 1,3
123 summarizes the demographics and statistics of these medical centers.

124 Overall, the BrainTumor46K dataset spans all 12 major brain tumor types and most of the subtypes defined by the 2021
125 WHO CNS5⁴. The detailed number of patients across the 12 major brain tumor types and their subtypes is shown in the right
126 panel of Fig. 1a. Specifically, BrainTumor46K includes gliomas, glioneuronal tumors, and neuronal tumors (GGN; $n = 12,740$);
127 meningiomas (MEN; $n = 8,320$); brain metastases (MET; $n = 2,437$); cranial and paraspinal nerve tumors (CPN; $n = 2,273$);
128 mesenchymal, non-meningothelial tumors (MNM; $n = 1,477$); hematolymphoid tumors (HEM; $n = 753$); tumors of the sellar
129 region (TSR; $n = 1,819$); choroid plexus tumors (CPT; $n = 378$); pineal tumors (PIN; $n = 417$); germ cell tumors (GCT; $n =$
130 449); embryonal tumors (EMB; $n = 518$); and melanocytic tumors (MEL; $n = 123$). Detailed demographic distributions for
131 each major tumor type are summarized in Table 1 and Extended Data Table 3, respectively. Tumor type acronyms are provided
132 in Supplementary Table 3.

133 All data obtained from participating medical institutions were obtained with ethics approval from their respective Institutional
134 Review Boards (IRBs). The requirement for informed consent was waived by the IRBs. All data were de-identified in compliance
135 with institutional policies and ethical standards.

136 Ready study for AI-augmented clinical assessments

137 To achieve a more rigorous and realistic evaluation of BrainVLM's clinical utility in augmenting neuroradiological diagnosis,
138 we developed a gold-standard test dataset comprising 123 patients covering 12 major WHO CNS5 brain tumor types. For each
139 case, readers received only the multi-parametric MRI scans (T1, T1c, T2, FLAIR) and patient metadata (age, gender); no clinical
140 history was provided. Diagnostic performance was systematically compared across three paradigms: (a) neuroradiologists-only
141 as a reader, (b) BrainVLM as an autonomous diagnostic agent, and (c) neuroradiologists augmented by BrainVLM as a reader.
142 In paradigm (c), BrainVLM assisted neuroradiologists by providing diagnoses with associated confidence scores and a clinical
143 rationale radiology report. Twelve neuroradiologists, stratified by expertise (5 juniors: 3-5 years; 4 seniors: 5-10 years; 3
144 experts: 10+ years), were recruited in the blinded crossover study. To minimize memorization effects in scenarios (a) and (c),
145 we reshuffled the order of the cases and modified their anonymized identifiers for the second read, and allowed a one-month
146 wash-out period between the two reads.

147 Prospective validation procedures

148 For the prospective study, we consecutively enrolled patients who received an initial diagnosis of brain lesion upon admission
149 at Xiangya Hospital between May 1, 2025 and July 31, 2025. Following brain MRI acquisition, clinical information and MRI
150 images were collected and preprocessed according to standardized protocols. Patients were excluded if they had a history of
151 prior brain surgery, previous brain radiotherapy or stereotactic radiosurgery, missing MRI images, or incomplete MRI protocols
152 (Fig. 3a). Eligible patients' data were prospectively predicted using BrainVLM prior to definitive clinical diagnosis. Following
153 patient discharge, BrainVLM outputs were validated against gold standard references: postoperative pathological findings for
154 patients who underwent surgery, or consensus discharge diagnoses established by a multidisciplinary panel of senior clinicians
155 through comprehensive review of radiological and clinical data for non-surgical cases. Finally, patients with non-tumoral
156 pathological findings were also excluded.

Development of AI foundation model

Developing an AI model for early brain tumor diagnosis necessitates the effective integration of presurgical multi-modal data, including multi-parametric MRI scans, patient metadata, and radiology reports. Current AI-based brain diagnosis approaches largely rely on the subjective interpretation of pure MRI data, sometimes supplemented with patient metadata^{29,30}. This neglects the rich, complementary information available in multi-modal datasets, potentially limiting the model's performance. We thus developed BrainVLM, a first-of-its-kind VLM that effectively synergizes multi-modal visual and textual information (Fig. 1b). Central to the BrainVLM framework are its clinician-trust focused innovations: 1) an integrated uncertainty quantification strategy explaining the degree of prediction reliability, and 2) a radiology report generation module elucidating the underlying clinical rationale. These advancements address the issues of overconfident errors and the opaque decision-making processes persistent in conventional AI-based diagnostic models. More details regarding the BrainVLM network architecture, uncertainty quantification strategy, and training strategies can be found in the Methods Section.

We trained BrainVLM on 40,663 individuals spanning all 12 WHO 2021 defined tumor types and nearly all the subtypes, with multi-modal data (MRI scans, demographics, and radiology reports) from diverse patient races, imaging equipment, and institutional protocols. These 40,663 individuals consist of 35,727 publicly available cases and 4,936 patients from the primary institutional cohort (Xiangya Hospital). We evaluated BrainVLM's diagnostic capabilities on 5,126 patients with pathologically confirmed brain tumors, including 3,797 held-out patients from the primary hospital and 1,329 patients from 11 independent hospitals (Fig. 1a).

Diagnostic performance of BrainVLM in primary test dataset

We first evaluated BrainVLM's diagnostic ability across all 12 WHO CNS5 major brain tumor types using the primary test dataset ($n = 3,797$). BrainVLM achieved a micro-averaged area under the ROC curve (AUC) of 0.91 (95% CI: 0.90–0.92; Fig. 1c) and a weighted-averaged F_1 score of 0.83 (95% CI: 0.82–0.84), with precision of 0.85 (95% CI: 0.84–0.86) and sensitivity of 0.82 (95% CI: 0.81–0.83; Fig. 1d). This exceeded expert neuroradiologists' assessments ($F_1 = 0.81$, 95% CI: 0.80–0.82; precision = 0.83, 95% CI: 0.82–0.84; sensitivity = 0.81, 95% CI: 0.80–0.82), with enhanced inter-rater agreement (Cohen's $\kappa = 0.76$, 95% CI: 0.75–0.77 vs. 0.74, 95% CI: 0.73–0.77). For common brain tumors (Fig. 2a), BrainVLM matched or surpassed expert-level neuroradiologists' performance: MEN ($F_1 = 0.91$ vs. 0.91), GGN ($F_1 = 0.83$ vs. 0.82), and CPN ($F_1 = 0.78$ vs. 0.74). In low-prevalence or imaging-ambiguous brain tumors, BrainVLM demonstrated consistent superiority; for example, HEM attained $F_1 = 0.64$ vs. 0.45, EMB reached $F_1 = 0.68$ vs. 0.58, and CPT achieved $F_1 = 0.54$ vs. 0.51. Confusion matrix analysis demonstrated substantial concordance between BrainVLM and human experts in diagnostic errors, with 83% overlap in Top 2 misclassified tumor categories (Fig. 2b). For example, both BrainVLM and neuroradiologists misclassified CPN tumors as MEN (6% vs. 5%) and 13% of MET cases as GGN compared to neuroradiologists' 14% error rate, while sharing similar challenges in GCT tumor differentiation (GCT→TSR/GGN: 17%/8% vs. 12%/13%).

Beyond clinical validation, we conducted technical benchmarking against three state-of-the-art VLMs: RadFM³¹ ($F_1 = 0.62$, 95% CI: 0.60–0.63), Merlin³² ($F_1 = 0.54$, 95% CI: 0.53–0.55), ChatGPT-4o³³ ($F_1 = 0.47$, 95% CI: 0.45–0.48), see Fig. 1d. BrainVLM showed $\geq 25\%$ improvement in key metrics, including rare subtypes (such as CPT, $F_1 = 0.54$ vs. 0.36; CPN $F_1 = 0.78$ vs. 0.40; EMB $F_1 = 0.68$ vs. 0.47) (see Extended Data Table 5), solidifying its technical superiority. Collectively, these results position BrainVLM as the first AI system to achieve radiologist-level performance on common brain tumors while advancing the detection of rare brain tumors, with error patterns mirroring clinical decision-making challenges.

Generalization to external dataset

To evaluate the generalizability of BrainVLM across institutions with heterogeneous data collection protocols, we validated BrainVLM on an external multicentric dataset. It comprises 1,329 patients from 11 independent medical institutions, spanning 11 major WHO-CNS5 tumor types (MEL data unavailable). BrainVLM demonstrated robust cross-institutional performance, achieving an AUC of 0.87 (95% CI: 0.86–0.89; Fig. 1c), and an F_1 score of 0.76 (95% CI: 0.73–0.78; precision = 0.77, 95% CI: 0.75–0.77; sensitivity = 0.76, 95% CI: 0.74–0.77; Fig. 1d). These results surpassed both radiologist consensus ($F_1 = 0.71$, precision = 0.72, sensitivity = 0.71), and state-of-the-art models (RadFM³¹: $F_1 = 0.43$, Merlin³²: $F_1 = 0.52$, ChatGPT-4o³³: $F_1 = 0.30$). Consistent with primary findings, BrainVLM matched or exceeded neuroradiologists across both common tumors and most of the rare tumor types: TSR ($F_1 = 0.87$ vs. 0.84), GGN ($F_1 = 0.80$ vs. 0.70), MET ($F_1 = 0.61$ vs. 0.52), CPN ($F_1 = 0.80$ vs. 0.74), EMB ($F_1 = 0.48$ vs. 0.42), and CPT ($F_1 = 0.29$ vs. 0.18). Confusion matrix analysis (Fig. 2b) revealed an 92% concordance between BrainVLM and human expert diagnostic patterns (e.g., MET→GGN: 22% vs. 17%). Notably, BrainVLM achieved these results using only standard MRI sequences (T1, T1c, T2, FLAIR) and demographic data, whereas neuroradiologists relied on supplemental advanced imaging (e.g., diffusion-weighted imaging, MR angiography, MR spectroscopy, and MR Perfusion) and patient histories in 56.1% of cases (Supplementary Table 1). Together, we demonstrated BrainVLM's unique capacity to deliver expert radiologist-level diagnostics across heterogeneous clinical environments without requiring advanced imaging or protocol harmonization.

Subcategories classification performance in GGN

GGN collectively represent a vital and diverse category of brain tumors with significant heterogeneity in therapeutic approaches and clinical prognoses. These tumors could be stratified into three subcategories according to the WHO CNS5 classification, including gliomas, glioneuronal tumors and neuronal tumors, and ependymal tumors. To extend the clinical utility of BrainVLM, we further evaluated the diagnostic performance of the model among these three subtypes, benchmarking against both neuroradiologists and state-of-the-art VLMs (RadFM³¹, Merlin³²). In this evaluation, the models were tasked with classifying brain tumors into 14 distinct categories, encompassing the 11 major WHO CNS5 tumor types and the three GGN subcategories. BrainVLM achieved F_1 scores of 0.78 for gliomas, 0.43 for glioneuronal tumors and neural tumors, and 0.30 for ependymal tumors (Fig. 2d) throughout the entire test cohort, demonstrating substantial improvements over both neuroradiologists consensus scores (0.75, 0.07, 0.24, respectively) and baseline models (RadFM: (0.60, 0.05, 0.09); Merlin: (0.53, 0.07, 0.14)). These results underscore BrainVLM's superior discriminative capability in fine-grained tumor categorization.

Preoperative molecular subgroup prediction in adult-type diffuse gliomas

Adult diffuse gliomas constitute the predominant malignant tumors affecting the central nervous system³⁴, comprising three main subtypes: IDH-mutant astrocytoma (IDH-m Astro), IDH-mutant oligodendroglioma (IDH-m ODG), and IDH-wildtype glioblastoma (IDH-w GBM)⁴. Accurate distinction among these subtypes is essential for optimizing treatment strategies and predicting patient outcomes^{35,36}. We finetuned BrainVLM using the publicly available adult-type diffuse gliomas cases from the BrainTumor46K dataset, and conducted threefold validation on the Xiangya primary dataset ($n = 494$; 333 IDH-w GBMs, 86 IDH-m Astros, 75 IDH-m ODGs). External validation was performed on an independent cohort ($n = 138$, collected from 11 medical centers; 102 IDH-wildtype GBM, 22 IDH-mutant astrocytoma, 14 IDH-mutant oligodendroglioma), with performance benchmarking against state-of-the-art VLMs (RadFM³¹, Merlin³²). In primary validation, BrainVLM achieved a micro-average AUC of 0.95 (95% CI: 0.93-0.96; Fig. 3e), surpassing Merlin³² (AUC 0.87, 95% CI: 0.86-0.87) and RadFM³¹ (AUC 0.83, 95%CI: 0.82-0.84). On external validation, BrainVLM maintained robust performance (AUC = 0.93), substantially exceeding Merlin³² (AUC = 0.83) and RadFM³¹ (AUC = 0.81). These results underscore the exceptional performance of BrainVLM for molecular subtyping, highlighting its potential as a versatile foundation model for brain tumor analysis. Further details on training and results are provided in the Supplementary.

Uncertainty-aware clinical decision support

Brain tumor diagnosis is a high-stakes clinical task, in which a misdiagnosis can lead to inappropriate or unnecessary treatments, prolonged patient suffering, and potentially life-threatening outcomes. However, LLMs and VLMs are susceptible to biases, frequent fabrications, and inaccuracies³⁷. Conventional VLMs may generate erroneous diagnoses with unwarranted high confidence or fail to provide uncertainty quantification³⁸, which can undermine clinicians' trust (Fig. 2c). To address this issue, we developed a consensus-driven strategy to augment BrainVLM with reliable uncertainty quantification (see method). Our analysis of 5,126 cases spanning both primary and external test datasets revealed that 73% of correct diagnoses ($n = 3,001/4,111$) were associated with high-confidence probabilities ($> 90\%$), while 70% of incorrectly diagnosed cases ($n = 711/1,015$) exhibited low confidence (50-85%) (Fig. 2c, middle). In contrast, conventional VLMs displayed low confidence in only 15% of incorrectly predicted cases ($n = 153/1,015$), as shown in the Vanilla BrainVLM results (Fig. 2c, left panel), where the consensus-driven strategy was not applied.

To mirror clinical workflows in clinically ambiguous cases, where neuroradiologists consider multiple diagnostic possibilities, we implemented a confidence-triggered supplementary diagnosis mechanism in BrainVLM (Extended Data Fig. 2d). When the confidence score for the primary diagnosis (Top 1) falls below a threshold (set as 75% in this study), the BrainVLM automatically presents the second-most probable diagnosis (Top 2) for radiologist review. On the primary dataset, this mechanism elevated Top 2 predictions from baseline AUC 0.91 to 0.95 (F_1 : 0.83→0.86), while external dataset performance improved from AUC 0.87 to 0.90 (F_1 : 0.76→0.78) (Fig. 1c). More specifically, BrainVLM surpassed neuroradiologists' Top 2 recommendations across 11 out of 12 tumor types in the primary dataset and 9 out of 11 tumor types in the external dataset, achieving comparable results in GCT brain tumor types but underperforming marginally in CPN (Extended Data Fig. 5).

Radiology report generation

Beyond its robust diagnostic performance across heterogeneous clinical settings, BrainVLM uniquely generates clinically interpretable radiology reports that provide a transparent rationale for each diagnosis. To quantify this capability, we compared BrainVLM-generated reports with those produced by several state-of-the-art VLMs, including ChatGPT-4o³³, Merlin³⁹, RadFM³¹. BrainVLM consistently outperformed all baselines across both primary and external test datasets (Fig. 3d). In the primary dataset, BrainVLM achieved superior BLEU-4⁴⁰, F1RadGraph-XL⁴¹, and RaTEScore⁴² scores (BLEU-4⁴⁰: 0.43 vs. 0.02–0.38; F1RadGraph-XL⁴¹: 0.57 vs. 0.15–0.46; RaTEScore⁴²: 0.75 vs. 0.47–0.63). Similarly, in the external dataset, BrainVLM demonstrated strong generalizability (BLEU-4⁴⁰: 0.35 vs. 0.02–0.29; F1RadGraph-XL⁴¹: 0.52 vs. 0.22–0.42;

RaTEScore⁴²: 0.69 vs. 0.42–0.58; Fig. 3d). Qualitative analysis revealed BrainVLM’s clinical precision, with generated reports matching neuroradiologists’ interpretations in lesion localization (e.g., tumor positioning), imaging feature characterization (e.g., T1/T2 signals, enhancement patterns), and identification of secondary findings (e.g., hydrocephalus, midline shift), see Fig. 3d and Extended Data Fig. 4.

AI-augmented clinical assessments

To achieve a more rigorous and realistic evaluation of BrainVLM’s clinical utility in augmenting neuroradiological diagnosis, we developed a gold-standard test dataset comprising 123 patients covering 12 major WHO CNS5 brain tumor types. For each case, readers received only the multi-parametric MRI scans (T1, T1c, T2, FLAIR) and patient metadata (age, gender); no clinical history was provided. Diagnostic performance was systematically compared across three paradigms: (a) neuroradiologists-only as a reader, (b) BrainVLM as an autonomous diagnostic agent, and (c) neuroradiologists augmented by BrainVLM as a reader. In paradigm (c), BrainVLM assisted neuroradiologists by providing diagnoses with associated confidence scores and a clinical rationale radiology report. Twelve neuroradiologists, stratified by expertise (5 juniors: 3–5 years; 4 seniors: 5–10 years; 3 experts: 10+ years), were recruited in the blinded crossover study. To minimize memorization effects in scenarios (a) and (c), we reshuffled the order of the cases and modified their anonymized identifiers for the second read, and allowed a one-month wash-out period between the two reads.

As an independent diagnostic agent, BrainVLM consistently outperformed junior and senior neuroradiologists across all brain tumor types and achieved comparable performance to that of expert neuroradiologists (Fig. 4b–d and Table 2). The AI-assisted neuroradiologists demonstrated a statistically significant 23.3% improvement in F_1 score across all experience levels compared to unaided assessments ($P < 0.001$), along with a 36.1% reduction in diagnostic time ($P < 0.001$). Among all brain tumor types, MEL, MNM, and EMB represent the greatest diagnostic challenges for neuroradiologists across experience levels, especially for junior and senior neuroradiologists. This may be attributed to their rarity and ambiguous or overlapping imaging characteristics. With AI assistance, F_1 scores in junior, senior, and expert groups could increase by 22%/32%/4% (MEL), 32%/24%/2% (MNM), and 53%/47%/20% (EMB), respectively, for these categories. This highlights BrainVLM’s ability to detect subtle MRI features often missed by human readers.

Overall, BrainVLM augmentation elevated junior and senior neuroradiologists’ diagnostic performance to near expert-level standards. Their mean F_1 scores improved from 0.53 (junior) and 0.57 (senior) to 0.71 and 0.74, respectively, which is only slightly below the expert neuroradiologists’ benchmark of 0.79 (Table 2). Notably, although BrainVLM achieved similar performance to that of expert neuroradiologists, AI-assisted experts yielded an encouraging improvement, with F_1 score improved from 0.79 to 0.86. These encouraging enhancements are driven by three synergistic components. First, BrainVLM and human experts exhibit complementary diagnostic strengths across tumor types. While expert neuroradiologists demonstrated superior performance in certain common tumors (e.g., MET and MEN), BrainVLM excelled in some rare types (e.g., EMB and CPT) (Table 2). This enables our BrainVLM to strategically augment human decision-making where needed most. Second, each of BrainVLM’s diagnoses is associated with a reliable uncertainty quantification, where most of the incorrect diagnoses are associated with low confidence scores. For example, when analyzing PIN — a category where BrainVLM underperformed relative to the expert — BrainVLM misclassified 3 of 4 PIN cases, yet all erroneous predictions were associated with low confidence (50–70%) (see Fig. 4a for the confidence distributions on these 123 cases). This uncertainty quantification strategy ensures that only high-confidence predictions are clinically actionable, while automatically triaging ambiguous cases to human experts for secondary review. Third, BrainVLM excels in generating comprehensive reports that provide neuroradiologists with actionable insights rather than isolated predictions, thereby improving diagnostic turnaround times. Collectively, these innovations and promising results establish BrainVLM as a transformative clinical tool that enhances diagnostic precision, reduces diagnostic time, and mitigates experience-dependent performance disparities in neuro-oncology.

Discussion

MRI serves as the cornerstone of presurgical diagnosis and monitoring of patients with brain tumors, leveraging its high spatial resolution, multiplanar reconstruction capabilities, and clinically validated multiparametric protocols⁴³. Manual interpretation of MRI scans persists as one of the most cognitively demanding and time-consuming tasks in radiology due to the overlapping imaging features across tumor types, inter-observer variability, and potential perceptual diagnostic errors. This diagnostic complexity is further compounded by the pathological diversity of brain tumors, which encompass 12 major categories and over 100 molecularly distinct subtypes. The extensive taxonomic spectrum of brain tumors creates persistent diagnostic challenges in radiology, particularly for rare subtypes requiring deliberate training efforts to mitigate^{44,45}.

To address these challenges, we developed BrainVLM, the first proof-of-concept vision-language foundation model capable of comprehensive classification across all 12 WHO CNS5-defined brain tumor types and three GNN subcategories. Trained on 40,043 patients spanning all 12 tumor types and most of the subtypes, BrainVLM achieved high diagnostic performance with robust generalizability, demonstrating AUCs of 0.91 on primary validation set and 0.87 on external validation set. It

outperformed board-certified neuroradiologists in sensitivity (0.82 and 0.76 vs. 0.81 and 0.71), precision (0.85 and 0.77 vs. 0.83 and 0.72), and F_1 scores (0.83 and 0.76 vs. 0.81 and 0.71), on primary and external validation sets, respectively. Furthermore, in subcategories specific evaluations (gliomas, glioneuronal tumors and neuronal tumors, and ependymal tumors) in GGN, BrainVLM demonstrated superior fine-grained classification compared to both expert neuroradiologists and state-of-the-art VLMs. In the molecular classification of adult-type diffuse gliomas, BrainVLM achieved a superior AUC on external validation (0.93) relative to state-of-the-art VLMs (0.81–0.83). In prospective real-world evaluation, BrainVLM demonstrated a statistically significant 4.7% higher F_1 score compared with neuroradiologists, supporting its clinical validity. These findings establish a foundation for future precise molecular classification prediction, which has become increasingly essential for the diagnosis and treatment of brain tumors^{46,47}.

The robust diagnostic performance of BrainVLM is primarily attributed to the unprecedented scale and diversity of its training dataset. To our knowledge, BrainTumor46K represents the largest and most comprehensive multi-modal presurgical dataset utilized for model development in neuro-oncology to date. Unlike current neuroimaging models constrained by public databases or single-center data with limited cases and limited tumor types¹⁷, BrainVLM was pretrained on an extensive corpus comprising public repositories and real-world clinical data. This extensive training enables the model to precisely interpret MR images of varying quality and acquisition parameters from diverse patient races, imaging equipment, and institutional protocols. Another key factor contributing to BrainVLM's performance is its efficient integration of multi-modal data, including MRI scans, patients' demographics, and radiological reports. This framework has enabled us to leverage critical clinical information such as age and radiology reports, leading to more accurate and comprehensive diagnoses.

AI-clinician collaboration holds significant promise in medical image interpretation^{37,48}. However, even when AI models exhibit robust diagnostic performance and generalizability, translating these capabilities into clinical practice necessitates overcoming a critical barrier: clinician trust in AI-driven decision-making. The effective integration of AI models into clinical workflows requires clinicians to both (1) understand the AI's decision rationale to inform their judgments, and (2) develop sufficient trust in the system's recommendations^{49,50}. To address these challenges, BrainVLM introduces two clinician-trust focused innovations: a consensus-driven uncertainty quantification strategy and an automated radiology report generation module. These advancements address the persistent issue of overconfident errors and opaque decision-making processes in conventional LLMs and VLMs⁵¹. In our evaluation, BrainVLM demonstrated clinically meaningful self-assessment capabilities, assigning high-confidence probabilities ($> 90\%$) to 70% of its correct diagnoses while appropriately tagging 74% of its incorrect diagnoses with low-confidence scores (50–85%). This self-awareness capability is particularly impactful in diagnostically challenging scenarios. This mechanism ensures only high-confidence predictions are clinically actionable, while automatically triaging ambiguous cases to human experts for secondary review. This approach also prevents clinicians from over-reliance on model outputs, thereby aligning with responsible clinical workflows in healthcare settings. Complementing this reliability mechanism, BrainVLM exhibits an exceptional capability to generate preliminary medical reports—a critical function for communicating diagnostic findings and guiding care across several specialties^{37,52}. These reports offer clinicians actionable insights beyond isolated predictions, enabling clinicians to process cases more efficiently, thereby improving turnaround times and expanding access to specialty-level reporting. Together, this dual functionality—diagnostic classification paired with reliability assessment and automated radiology report generation—positions BrainVLM as a paradigm shift in AI-assisted neuro-oncology. The clinical validation through a multi-reader study with 12 neuroradiologists across expertise levels further confirmed BrainVLM's trust-building potential.

In clinical practice, incomplete MRI acquisition (e.g., missing T1 contrast-enhanced or FLAIR) frequently occurs due to insufficient clinician awareness or contraindications. While conventional multi-sequence-trained models exhibit considerable performance degradation under such suboptimal imaging conditions^{53,54}, BrainVLM maintains diagnostic performance comparable to the full-sequence baseline in scenarios where T2, FLAIR, or T1 sequences are missing. A notable exception is the absence of T1c sequences, where we observe an 11.4% decline in F_1 score (Extended Data Table 4), highlighting the critical role of post-contrast imaging in brain tumor assessment.

BrainVLM's clinical validity is further strengthened by its unique training paradigm, which utilizes raw, minimally processed MR images, without skull stripping or segmentation. Unlike conventional neuroimaging studies, which typically require preprocessing steps such as skull stripping^{15,22,55–58}, BrainVLM directly analyzes raw institutional data, with only slice-level resampling applied. This minimal preprocessing preserves the complete informational content of the original MRI data, making BrainVLM particularly well-suited for our comprehensive brain tumor characterization study. While conventional skull-stripping may aid certain tumor analyses, it is not universally appropriate. For example, some brain lesions are located in the pia mater or invade the cranial vault (see Supplementary Fig. 2), where the skull stripping can inadvertently remove important anatomical information in these regions. Similarly, segmentation, another popularly used preprocessing strategy in current diagnostic MRI foundation models^{17,59,60}, may introduce considerable inter-reader variability on both automated and manual segmentation workflows. This inconsistency may result in the omission of subtle yet significant diagnostic features located at tumor margins or within the surrounding parenchyma⁶¹. Collectively, BrainVLM's ability to deliver robust diagnostic

performance without preprocessing workflows (e.g., skull stripping) or segmentation requirements, combined with its resilience to missing MRI sequences, establishes its clinical efficacy as an adaptable and practical solution for real-world neuroimaging challenges.

Several limitations exist in our study. First, BrainVLM was developed using the BrainTumor46K dataset and validated across 12 medical centers. While online datasets were collected globally, the real-world diagnostic applications were exclusively from Xiangya Hospital (a hospital in China), and the external test cohort comprised solely East Asian patients. Further validation studies are warranted to comprehensively assess the model’s generalizability across diverse ethnic populations. Second, although BrainVLM demonstrated exceptional performance in classifying brain tumors across 12 general categories and three glioma-related subcategories, the current diagnostic framework does not yet enable precise differentiation of molecular tumor categories. As novel therapeutic approaches increasingly target specific molecular characteristics^{62,63}, there exists a critical need to extend the model’s training to encompass finer-grained molecular pathological classifications that can directly inform personalized treatment selection and prognostic assessment in neuro-oncology. Finally, while patient demographics are incorporated into BrainVLM, integrating multimodal clinical data, such as longitudinal symptom progression or treatment response patterns, could further enhance diagnostic precision by aligning the model with real-world clinical decision-making^{6,15}.

Methods

Construction of BrainTumor46K dataset

We assembled the BrainTumor46K dataset through systematic curation from 36 online sources and 12 collaborating medical centers. To guide the search and data collection processes, we first established a comprehensive set of brain tumor keywords, including 12 major brain tumor types and all subtypes, along with their synonyms and related terms. Next, we leveraged retrieval-augmented generation (RAG) and prompt engineering techniques to design an automated preprocessing pipeline, see Extended Data Fig 1. This pipeline transformed the curated raw datasets into structured formats of {*Patient_ID*: multi-parametric 3D MRI volumes (or 2D MRI slice), pathological diagnosis label, *optional* radiology report (or text description)} for model training. We also expanded our datasets by incorporating data from healthy individuals. For the multi-parametric MRI scans, we considered T1-weighted (T1), T1 contrast-enhanced (T1c), T2-weighted (T2), and T2-FLAIR (FLAIR), which are commonly used in brain tumor diagnosis.

Dataset construction from online repositories

We compiled our dataset from diverse online repositories, categorized based on their inherent structure and the required preprocessing complexity. The first category well-structured sources of medical imaging data, such as established repositories (e.g., BraTS2023^{21,22,64–67}, TCIA^{68–81}, Kaggle^{82–85}, Ctiscus⁸⁶) as well as specialized medical platforms like Radiopaedia⁸⁷. This category encompasses sources that provide readily accessible, structured medical imaging data (including 3D multi-parametric MRI volumes or 2D slices), which are typically paired with pathological labels. Depending on the source, additional information, such as patient metadata or radiology reports, may also be available. The second category involves non-dedicated imaging sources requiring advanced figure/text extraction, such as PubMed Central scientific archives⁸⁸. These platforms, not originally designed for medical imaging storage, necessitated specialized pipelines for isolating figures/text from heterogeneous formats (e.g., publication figures, video stills) and reconciling fragmented clinical annotations. In the following, we detail the distinct data pre-processing pipelines developed for each category.

Pre-processing pipeline for structured repositories and pre-curated datasets. Datasets in this category are highly heterogeneous, encompassing various sources with differing image formats, annotations, and metadata. These datasets range from simplified collections (where each patient is represented by a single 2D MRI slice with a slice-level pathological label) to comprehensive datasets containing multi-parametric 3D MRI sequences, tumor segmentation masks, patient metadata, and pathological labels. For datasets providing only tumor classification labels, we processed the data into a standardized structure of the form: {*Patient_ID*: 3D MRI volumes (or 2D MRI slice), pathological label}. The resulting standardized data were subsequently used for tumor classification tasks. For datasets containing 3D multi-parametric MRI scans (or 2D MRI slice), segmentation masks, and metadata (e.g., BraTS^{21,22,64–67}, TCIA^{68–81} and Kaggle datasets^{82,83}), we first applied the Brainnetome Atlas⁸⁹ to localize functional brain regions based on the tumor masks. We then leveraged all available metadata (including imaging modality, gender, diagnostic label, and tumor description) to generate comprehensive medical reports via ChatGPT-4o prompted with RAG techniques (see Extended Data Fig. 1a1). For certain web-sourced datasets, such as Radiopaedia⁸⁷, which include multi-parametric MRI scans and accompanying radiology reports, we removed irrelevant information and reformatted the data into a consistent structure: {*Patient_ID*: 3D MRI volumes, pathological label, radiology report}.

This comprehensive pre-processing pipeline enabled the assembly of a diverse cohort comprising (see Extended Data Table 2): 1) 6,957 brain tumor patients with 3D multi-parametric MRI scans, radiology reports, and associated pathological labels;

2) 876 brain tumor patients with 3D multi-parametric MRI scans and pathological labels only; 3) 233 patients with 2D MRI slice-text description; and 4) 3,178 healthy control patients with multi-parametric 3D MRI scans. Apart from the datasets detailed above, this preprocessing pipeline also incorporated subjects from additional repositories, such as Br35H⁸², Brain Tumor Classification Dataset⁸⁵, and Brain Tumor MRI Images 44 Classes⁸⁴. However, as these repositories only contain 2D MRI slices with associated pathological labels and lack patient identifiers, they were not included in our dataset summary here (see Extended Data Table 2).

Pre-processing pipeline for unstructured web-based resources. Web-crawled brain tumor data from platforms such as PubMed Central⁸⁸ and ImageCLEF⁹⁰ often contain noisy 2D MRI scans and misaligned or inconsistent text descriptions. To address these challenges, we developed a comprehensive data preprocessing pipeline that transforms raw, heterogeneous web data into a structured format: {*Patient_ID*: 2D MRI slice, text description, pathological label}, see Extended Data Fig. 1a2.

For scientific articles from PubMed Central, we systematically retrieved 2D brain MRI slices by querying predefined brain tumor keywords, associating each image with both “golden captions” and inline textual mentions as candidate descriptions. However, these figures frequently contain composite images with multiple subfigures and inconsistent captioning. To resolve this, we developed a three-stage refinement protocol: Firstly, we applied a Faster R-CNN object detector⁹¹, trained on the MedICaT dataset⁹², to perform subfigure separation and irrelevant images filtering. Secondly, we employed ChatGPT-4o with specialized prompts for accurate subcaption separation. Finally, we established a precise one-to-one subfigure-caption correspondence through bounding box coordinate alignment⁹³. ImageCLEF data required minimal processing, with only irrelevant images filtered, as slice-text pairs were directly utilized.

In total, our curated dataset includes 24,333 unique subjects: 23,849 from PubMed Central (116,296 2D MRI slice-text pairs), and 484 from ImageCLEF (4,631 2D MRI slice-text pairs).

Dataset construction from medical institutions

To address the scarcity of rare brain tumor types/subtypes and the limited availability of expert-curated radiology reports in online repositories, we curated data from twelve collaborating hospitals (Extended Data Table 1). For each patient with a pathologically confirmed diagnosis, we collected de-identified multi-parametric MRI scans (T1, T1c, T2, FLAIR) that are commonly used for CNS tumor diagnosis, along with de-identified radiology reports and associated pathological diagnoses. In total, we gathered multi-modal data from 8,733 patients at the primary Xiangya Hospital and 1,329 patients from 11 other independent institutions. Demographic and statistical summaries of these medical centers are provided in Fig. 1 and Extended Data Table 1.

The original institutional radiology reports were in Chinese and often included descriptions of other modalities (e.g., Magnetic Resonance Angiography (MRA), Magnetic Resonance Spectroscopy (MRS), Blood Oxygen Level Dependent Imaging (BOLD) signals; Supplementary Table 1). Thus, we implemented a four-stage standardization protocol to process these reports (Extended Data Fig. 1c). Firstly, we used regular-expression filters to eliminate non-T1/T1c/T2/FLAIR content. Secondly, we developed a clinician-validated bilingual lexicon (Chinese-English) to accurately map medical terms, translating terms such as “T2 长信号” to “T2 Hyperintense” and “脑实质” to “brain parenchyma”. Thirdly, leveraging this lexicon as a knowledge base, we utilized RAG techniques and prompt engineering strategies through the advanced capabilities of ChatGPT-4o³³, to facilitate the translation of the reports from Chinese to English. Finally, the translated reports are thoroughly reviewed and refined by board-certified neuroradiologists to ensure their accuracy.

Ethics approval for datasets from medical institutions

All data obtained from participating private medical institutions were obtained with ethics approval from their respective Institutional Review Boards (IRBs). This includes Xiangya Hospital, Changde First People’s Hospital, Shanghai Tongji Hospital, University of South China Second Hospital, Shenzhen Second People’s Hospital, the Third Xiangya Hospital, Jiangxi Provincial People’s Hospital, Chongqing Traditional Chinese Medicine Hospital, the First Affiliated Hospital of Nanchang University, Hunan Provincial Children’s Hospital, the First Affiliated Hospital of Lanzhou University, and the Second Affiliated Hospital of Nanchang University. The requirement for informed consent was waived by the IRBs. All data were de-identified in compliance with institutional policies and ethical standards.

Estimation of total subject count from online resources

Accurately estimating subject-level statistics from the aggregated 36 online resources is challenging due to the inconsistent presence or absence of unique patient identifiers. For resources that do provide explicit patient identifiers, such as TCIA and BraTS, patient counts were directly extracted. In contrast, for web-crawled resources like PubMed Central⁸⁸, and ImageCLEF⁹⁰, we conservatively assigned one unique patient per composite figure. This methodology likely results in an underestimate, as subfigures within composites may pertain to distinct cases, and images from the same video may originate from different patients. For certain directly downloaded datasets, including the Br35H⁸², Brain Tumor Classification Dataset⁸⁵, and Brain

Tumor MRI Images 44 Classes, which lack patient identifiers, we excluded these from our subject count estimation. The detailed number of patients and their associated multi-modal data in each online resource are illustrated in Extended Data Table 2.

Development of BrainVLM for tumor classification and report generation

Developing an AI model for early brain tumor diagnosis necessitates the effective integration of presurgical multi-modal data, including multi-parametric MRI scans, patient metadata, and radiology reports. Current AI-based brain diagnostic approaches largely rely on the subjective interpretation of pure MRI data, sometimes supplemented with patient metadata^{29,30}. This neglects the rich, complementary information available in multi-modal datasets, potentially limiting the model’s performance. Besides, the current vision language foundation models (VLMs⁹⁴) face challenges as the input length increases with the quantity and resolution of images. This is particularly alarming in brain tumor studies, which require the analysis of high-resolution (HR), multi-parametric MRI scans (T1, T1c, T2, FLAIR) from various views (axial, coronal, sagittal).

To address this, we developed BrainVLM, a first-of-its-kind VLM that effectively synergizes multi-modal medical visual and textual information. It tackles the challenges through optimization across three aspects: (1) Innovative multi-modal model architecture: Incorporating anatomical detail-aware vision encoder and hybrid causal-full attention strategies into a multi-modal network to handle complex multi-modal data efficiently. (2) Clinical-motivated data standardized: Developing a standardized dataset processing protocol that emulates clinical practices and incorporates special markers to differentiate different visual data. (3) Progressive three-stage training strategy: Using a progressive training approach to incrementally enhance the model’s multi-modal processing capabilities.

Multi-modal model architecture

As shown in Fig. 1b, BrainVLM consists of three core components: (1) an anatomical detail-aware vision encoder for efficient, fine-grained image feature extraction, (2) a 2-layer Multilayer Perceptron (MLP) projector that bridges the image encoder with the LLM, and (3) a hybrid causal-full attention-based LLM for reasoning and understanding the multi-modal information.

Anatomical detail-aware vision encoder. We employed the vision backbone of BioMedCLIP⁹⁵ as the basic visual encoder E . This model constitutes a Vision Transformer (ViT) pre-trained on a large-scale 2D medical imaging dataset using vision-language contrastive learning⁹⁶. The standard ViT of original BioMedCLIP accepts only 224×224 input dimensions and cannot directly process high-resolution (HR) MRI slices (typically 400–500 pixels in width/height). Direct downsampling risks losing critical diagnostic details such as tumor margins and signal heterogeneity, while splitting the image into several sub-images for individual encoding results in excessive visual tokens for LLMs. This is computationally prohibitive due to the transformer’s quadratic complexity. We therefore designed an anatomical detail-aware vision encoder to capture finer anatomical details while maintaining computational efficiency.

Our designed visual encoder incorporated two key components: mixed-resolution visual encoding and attention-based HR injection. Let I denote an input MRI scan with dimensions $N \times 3 \times w_o \times h_o$, where N represents the number of slices (e.g., $N = 1$ for single-slice MRI; $N = 32$ for a 3D MRI volume with 32 MRI slices), and $w_o \times h_o$ specifies the original spatial resolution. We first generated two resized versions: 1) a low-resolution (LR) input I^l scaled down to $N \times 3 \times 224 \times 224$, and 2) an HR input I^h scaled up to size $N \times 3 \times 448 \times 448$. The LR input was encoded through the visual encoder E to generate global LR tokens $z^l = E(I^l) \in \mathbb{R}^{N \times 768}$. Here, z^l is extracted from the final layer’s [CLS] token of the ViT, capturing global patterns.

Next, I^h was partitioned into four non-overlapping HR sub-inputs $\{I_k^h\}_{k=1}^4$ using a grid method⁹⁷, where k indexes the HR sub-inputs. Each HR sub-input was resized to $N \times 3 \times 224 \times 224$ and independently encoded by E to generate HR visual tokens as $z^{h_k} = E(I_k^h)$. Unlike the global [CLS] token extraction used for the LR input, we retained all spatial patch tokens (excluding [CLS]) from the ViT’s final layer when processing HR sub-inputs. This yields features $z^{h_k} \in \mathbb{R}^{N \times 196 \times 768}$ that encode fine-grained anatomical details. The final combined HR representation is denoted as $z^h = \{z^{h_k}\}_{k=1}^4 \in \mathbb{R}^{4 \times N \times 196 \times 768}$.

Finally, we injected z^h into z^l via an LR-HR cross-attention module, to obtain HR-informed LR tokens z^{hl} . Here, z^l served as query Q , $\{z^{h_k}\}_{k=1}^4$ served as the context (key K and value V), resulting the final $z^{hl} \in \mathbb{R}^{N \times 768}$. This approach merged global context z^l with anatomical details from z^h , while maintaining computational efficiency.

For multi-parametric MRI based brain tumor diagnosis (e.g., T1, T1c, T2, FLAIR), each 3D MRI sequence was processed independently by our encoder to extract modality-specific features, denoted as $z_m^{hl} \in \mathbb{R}^{N \times 768}$. These features were then concatenated along the slice dimension: $z^{\text{comb}} = \parallel_{m=1}^M z_m^{hl} \in \mathbb{R}^{(M \cdot N) \times 768}$, where M denotes the number of MRI sequences. For example, a 5-sequence protocol (e.g., T1 axial, T1c axial, T1c sagittal, T2 axial, FLAIR axial) with $N = 32$ slices per sequence yielded a combined representation $z^{\text{comb}} \in \mathbb{R}^{160 \times 768}$, integrating complementary diagnostic information across sequences.

MLP projector. The combined feature representation z^{comb} was mapped into the LLM language space via a 2-layer MLP projector, converting visual information into a format digestible for LLM. This transformation bridged the modality gap by converting anatomical visual patterns into a sequential token format compatible with linguistic processing.

Hybrid causal-full attention based LLM. The MLP-projected visual embeddings were combined with patient metadata (e.g., age and gender) and textual instructions to form a multimodal prompt. This prompt was then processed by our adapted LLM for tumor classification, uncertainty quantification, and report generation. We utilized the pre-trained Llama-3.1-8B-Instruct⁹⁸ model as our base LLM, which provides a mainstream language backbone with grouped-query attention and open-source adaptability. Traditional causal attention in Llama-3.1-8B-Instruct restricts tokens to preceding positions only, which is suboptimal for MRI interpretation⁹⁹. Therefore, we implemented a hybrid attention strategy following previous works^{100,101}. Specifically, full attention was applied to image tokens for capturing complex visual interdependencies within the same 3D MRI sequence or 2D MRI slice, while causal attention was retained for sequential tokens to preserve data structure. This hybrid architecture enabled the model to simultaneously capture cross-modal visual relationships and maintain linguistic coherence.

Clinical-motivated data standardization protocol

Language data in LLM is typically uniform and consistent, while our task involves more complex visual data. Randomly feeding unprocessed neuroimaging inputs from different modalities and views into BrainVLM complicates the learning process and impedes model convergence. Thus, we proposed a standardized dataset processing protocol that emulates clinical practices through the following three key strategies: 1) volumetric slice-count harmonization to ensure uniform slice dimensions, 2) clinical MRI sequence integration mirroring radiological workflows, and 3) special markers to differentiate multi-modal data and task-driven prompt design to differentiate different tasks. This protocol reduced learning ambiguity while preserving diagnostically critical features essential for brain tumor analysis.

MRI preprocessing via volumetric slice-count harmonization. The 3D MRI sequences used in our study typically exhibit substantial variability in the number of acquired slices. This variation occurs both across scanners/institutions (ranging from dozens to hundreds of slices per sequence) and between sequences for the same subject (e.g., 20 slices in native T1 vs. 40 in T1c sequences). This substantial slice variation significantly complicates model training. We therefore standardized each 3D MRI sequence to 32 slices using cubic spline interpolation (implemented via SimpleITK¹⁰²), ensuring smooth volumetric resampling along the slice direction. Critically, unlike conventional neuroimaging pipelines that typically involve extensive preprocessing — such as resampling to isotropic voxels and skull stripping — BrainVLM performs *only* this volumetric slice-count harmonization. This minimal preprocessing preserves the complete information content of the original MRI data, making it particularly well-suited for our comprehensive brain tumor characterization study. While conventional skull-stripping may aid certain tumor analyses, it is not universally appropriate. Crucially, lesions located in regions like the pia mater or invading the cranial vault can be inadvertently removed during skull stripping, leading to the loss of critical anatomical and pathological information — a risk our approach deliberately avoids (Supplementary Fig. 2).

Clinical MRI sequence integration. Emulating the clinical workflow for MRI-based tumor identification is critical for aligning machine learning models with diagnostic practice, thereby improving lesion identification and characterization^{103,104}. In clinical practice, comprehensive brain tumor assessment often initiates with acquiring axial T1 and T2 sequences, *optionally* complemented by axial FLAIR imaging. Following intravenous administration of gadolinium-based contrast agents, multiple-plane T1c (axial, sagittal, and coronal views) are obtained to evaluate enhancement patterns and blood-brain barrier disruption. Based on initial findings or clinical suspicion, additional sequences or planes (e.g., coronal or sagittal T1/T2 for sellar region lesions) may be acquired. During interpretation, neuroradiologists typically begin by examining T2 or FLAIR images to identify potential lesions. They then use T1c images to detect small lesions and assess tumor enhancement, comparing these with T1 images of the same view for confirmation.

To emulate this clinical workflow and streamline model learning, we organized the MRI data to reflect both the clinical acquisition and interpretation workflows. Specifically, our integration strategy prioritizes clinically meaningful comparisons: the primary input consisted of paired native T1 and T1c scans acquired in the same plane, facilitating direct assessment of lesion enhancement. This core pairing was supplemented by representative T2 and FLAIR scans, as well as an additional T1c scan obtained from a different imaging plane than the primary T1/T1c pair. For example, if the primary T1/T1c pair was axial, we included an additional T1c scan from the coronal or sagittal plane to provide multiplanar context. When specific sequences or planes were unavailable, alternative available sequences were included to maximize diagnostic coverage. In summary, BrainVLM utilized five core 3D MRI sequences as visual input—T1 (axial or another view), T1c from the same view as T1, T2 (axial or another view), FLAIR (axial or another view), and an additional T1c from a different view than T1. This structured yet flexible approach maintained comprehensive diagnostic information while faithfully reflecting real-world clinical reasoning. Further details regarding the processing of cases with more than five MRI sequences are provided in the following Model Inference Section .

Special tokens for data type differentiation and task-driven prompt design. BrainVLM processed heterogeneous inputs spanning 2D MRI slices, 3D MRI volumes, and free-text patient metadata (e.g., age, sex), while concurrently addressing

distinct task types including classification, report generation, and reliability assessment. To enable the model to distinguish between these data types and tasks, we implemented the following specialized tokens and task-driven prompts. Specifically, to help the model distinguish different data types, we designed special tokens: 2D single image slices were enclosed with '<image>' and '</image>', while 3D MRI volumes used '<vid>' and '</vid>' tags. A special token '<s>' was inserted between volumes to indicate the different sequence's input. We also introduced specialized tumor tokens (e.g., '<class_0>' for MET) to encode the original tumor names for classification. Concurrently, to enhance the model's task-specific learning capabilities, we designed customized prompts that explicitly inform the model of the input structure and expected objectives (see Extended Data Table 4). Specifically, the finalized prompt that was input into BrainVLM integrates three components: (1) predefined special visual tokens (e.g., '<vid>', '<image>') to structure the imaging inputs; (2) patient metadata; and (3) task-specific instructions that explicitly define the model's objective for each task (e.g., generating a clinical description for patients in the radiology report generation task or making a self-assessment confidence score for the reliability task). These special tokens and unified prompt design ensure robust alignment between input interpretation and clinical task requirements, enhancing the model's interpretative accuracy and adaptability to diverse diagnostic objectives.

Three-stage progressive training strategy of BrainVLM

Our curated BrainTumor46K dataset comprises three distinct data types: 54.0% 2D MRI slices paired with text descriptions, 37.2% multi-parametric 3D MRI volumes with radiology reports, and 8.8% 3D MRI sequences annotated solely with diagnostic labels. Direct end-to-end training on this heterogeneous data composition may overwhelm models with conflicting objectives. The 2D isolated MRI slices lack volumetric context essential for tumor characterization, while complete volumetric inputs demand foundational slice-level recognition capabilities that 2D training establishes. To address these challenges, we implemented a curriculum learning paradigm with three progressive training stages. This strategy enables the model to progressively interpret complex visual patterns, starting from simple 2D slices analysis and advancing to more intricate 3D volumetric data interpretation (see Extended Data Fig. 2a).

Training stage 1: 2D MRI slice-text representation learning. Volumetric tumor analysis fundamentally requires slice-level recognition capabilities, a skill radiologists cultivate through detailed examination of individual MRI slices. To establish this foundation while maximizing BrainTumor46K's heterogeneous resources, we initiated BrainVLM's training with a focus on 2D MRI slice-level representation learning. During this phase, we optimized only the MLP projector and the LR-HR cross-attention module within our anatomical detail-aware encoder, keeping the vision encoder and LLM module frozen to maintain stability. This stage focused on training 2D MRI slice-text alignment and utilized two main sources: (i) all available 2D MRI slice-text pairs in BrainTumor46K, and (ii) strategically extracted tumor slices from 3D MRI volumetric sequences, paired with AI-generated slice-level text descriptions. For public datasets with tumor masks, such as BraTS, we extracted slices with more than 5% tumor occupancy and generated clinically validated text descriptions using RAG, refined by ChatGPT-4o. For institutional scans, neuroradiologists manually selected pathologically significant slices from each 3D MRI sequence, and the corresponding slice-level reports were crafted using RAG refined by ChatGPT-4o. Finally, this meticulous process resulted in 10,996,702 curated 2D MRI slice-text pairs. This 2D MRI slice-level representation learning approach enabled BrainVLM to establish fundamental tumor recognition capabilities, laying a solid foundation for subsequent advanced 3D volumetric analysis.

Training stage 2: Volumetric context integration via hybrid 2D-3D training. In this phase, we aimed to transfer the learned 2D representation capabilities into the tomographic 3D MRI data space, facilitating BrainVLM's 3D tumor characterization capability. We utilized all available 2D and 3D training datasets from BrainTumor46K and designed a dynamic learning strategy interleaving 2D slice-level and 3D volumetric tasks during training. Crucially, at each iteration, we stochastically sampled training instances with 30% probability allocated to 2D tasks (leveraging isolated slices for tumor description and slice-level diagnosis) and 70% probability to 3D tasks (utilizing full sequences for volumetric report generation, classification, and diagnosis). During this stage, the MLP projector, LR-HR cross-attention module, and Low-Rank Adaptation (LoRA¹⁰⁵) parameters in the LLM model were trained, while the vision encoder and other parameters in the LLM model remained frozen. This transitional stage ensures that the model develops a cohesive understanding of tumor characteristics across different dimensions.

Training stage 3: Comprehensive 3D volumetric instruction tuning. In the final stage, we focused on leveraging the full high-quality 3D data available in the BrainTumor46K dataset to enhance BrainVLM's advanced volumetric analysis capabilities. The projector, LR-HR cross-attention module and LoRA parameters in the LLM model were trained, while all other parameters remained frozen.

Loss function

We introduced dual-objective losses to unleash BrainVLM’s generative and discriminative capabilities across clinical tasks, including free-text generation and dedicated tumor classification. Specifically, we first employed a standard causal language modeling loss $\mathcal{L}_{\text{LM}}$, which enables the model to follow natural language instructions and generate coherent diagnostic texts. Given a token sequence $(w_i^{(1)}, w_i^{(2)}, \dots, w_i^{(T)})$ in a training sample x_i , this loss is formulated as:

$$\mathcal{L}_{\text{LM}} = - \sum_{t=1}^T \log P(w_i^{(t)} | w_i^{(1)}, w_i^{(2)}, \dots, w_i^{(t-1)}; \theta), \quad (1)$$

where $P(w_i^{(t)} | w_i^{(1)}, w_i^{(2)}, \dots, w_i^{(t-1)}; \theta)$ denotes the predicted probability of token $w_i^{(t)}$ given the preceding tokens. θ denotes the model parameters. $t = 1, \dots, T$ denotes the tokens in the sequence. This objective encourages contextual understanding and fluent text generation by minimizing the negative log-likelihood of ground-truth tokens.

Furthermore, to enhance the tumor-specific discrimination, we introduced an auxiliary classification loss $\mathcal{L}_{\text{CL}}$ to perceive fine-grained medical patterns. We added special tokens to encode the original tumor label (such as `<tumor_3>` token for Glioma), and employed a classification loss based on these tokens. This loss is computed as:

$$\mathcal{L}_{\text{CL}} = - \sum_{c=1}^{14} y_c^{(i)} \log \hat{y}_c^{(i)}, \quad (2)$$

where $y_c^{(i)}$ denotes the ground-truth label for the c -th tumor class, and $\hat{y}_c^{(i)}$ is the predicted probability. This loss enables fine-grained recognition across 14 tumor categories, including 11 major brain tumor types in WHO-CNS5 and 3 GNN subcategories.

The overall objective combines both losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \alpha \mathcal{L}_{\text{CL}}, \quad (3)$$

where α is a tunable coefficient balancing language generation and tumor classification. This joint optimization framework enables BrainVLM to effectively capture both general linguistic semantics and task-specific diagnostic cues.

Data augmentation

Brain MRI data exhibited domain shifts driven by patient-specific factors (e.g., age, gender, physique) and MRI hardware variability¹⁰⁶, resulting in significant intra-modality variation. We applied tailored augmentation strategies for both 2D and 3D MRI data to improve BrainVLM’s generalization across diverse clinical settings. For 2D MRI slices, we adopted standard augmentation techniques¹⁰⁷, including random flipping, cropping, and intensity transformations. For 3D MRI sequences, we first identified key challenges such as noise artifacts, intensity inhomogeneity, and motion-induced blurring (examples shown in Supplementary Fig. 1). Then, to improve robustness against these effects, each 3D training sample was augmented with: (1) random rotation within ± 60 degrees, and (2) intensity variation, motion blur, or random noise, each applied with 30% probability.

The uncertainty quantification strategy in BrainVLM

Brain tumor diagnosis is a high-stakes task, where a misdiagnosis can lead to inappropriate treatments, prolonged patient suffering, and potentially fatal outcomes. However, conventional VLMs may generate erroneous predictions with high confidence³⁸, undermining clinical trust and limiting practical utility. As illustrated in the left panel of Fig. 2c in the main manuscript, our vanilla BrainVLM model (without reliability mechanisms) also exhibited this concerning behavior, frequently assigning high confidence scores to erroneous predictions. To address this critical limitation, we presented an uncertainty quantification strategy that calibrates prediction confidence in BrainVLM: accurate diagnoses receive low uncertainty scores (high confidence score), while erroneous predictions are assigned high uncertainty (low confidence score). This ensures only high-confidence, reliable predictions are acted upon.

Existing uncertainty quantification approaches for VLMs and LLMs rely on multiple sampling strategies (e.g., Best-of-N) that incur substantial computational overhead regardless of question complexity¹⁰⁸. While adaptive sampling techniques attempt to reduce costs through early-stopping heuristics, they remain constrained by manually designed rules that lack generalizability across models and tasks¹⁰⁹. We introduce a consensus-driven confidence calibration strategy that directly transforms model disagreement into calibrated confidence signals, eliminating the need for heuristic rules or fixed sampling costs. Our approach features two key innovations: (1) Consensus-driven reliability dataset construction, where a reliability dataset is automatically constructed using agreement statistics from multiple model instances via consensus-driven clustering; (2) Confidence-aware reliable BrainVLM finetuning: Distillation of consensus-derived reliability knowledge into our BrainVLM framework via parameter-efficient fine-tuning. We refer to the baseline BrainVLM without uncertainty quantification as the vanilla BrainVLM, while our enhanced version incorporating the proposed uncertainty quantification strategy is termed reliable BrainVLM.

Consensus-driven reliability dataset construction

To create a high-quality reliability dataset for confidence calibration, we leveraged the natural training dynamics of BrainVLM. Specifically, when our best-performing model was near convergence, we sampled N model checkpoints at different training iterations to create an ensemble of N vanilla BrainVLM variants. For each training sample x_i with pathological label $y_{\text{true}}^{(i)}$, we independently processed x_i through all N models to obtain N diagnostic predictions $\{\hat{y}_i^{(1)}, \dots, \hat{y}_i^{(N)}\}$ per sample (see Extended Data Fig. 2b). These responses were then clustered by predicted diagnoses, with the largest cluster defining the consensus label $y_{\text{cons}}^{(i)}$. Formally, the consensus label is computed as:

$$y_{\text{cons}}^{(i)} = \arg \max_{c \in \mathcal{C}} \sum_{k=1}^N \mathbb{I}(\hat{y}_i^{(k)} = c), \quad (4)$$

where $\mathcal{C}$ denotes the set of all possible diagnostic classes and $\mathbb{I}$ indicates the indicator function. This consensus label represents the prediction with the strongest inter-model agreement. We then computed a confidence score $s_{\text{con}}^{(i)}$ for each sample x_i :

$$s_{\text{con}}^{(i)} = \frac{N - N_w^{(i)}}{N}, \quad (5)$$

where $N_w^{(i)} = \sum_{k=1}^N \mathbb{I}(\hat{y}_i^{(k)} \neq y_{\text{true}}^{(i)})$, which indicates the number of incorrect predictions among the N models. A high N_w implies greater disagreement, resulting in a lower confidence score, while lower N_w yields higher confidence scores.

For each sample $(x_i, y_{\text{true}}^{(i)})$, the above process yields per-sample supervision tuples $(x_i, y_{\text{cons}}^{(i)}, s_{\text{con}}^{(i)})$, where $y_{\text{cons}}^{(i)}$ represents the ensemble-aggregated prediction, and $s_{\text{con}}^{(i)}$ quantifies reliability based on inter-model consensus strength $y_{\text{cons}}^{(i)}$ and clinical validity alignment with $y_{\text{true}}^{(i)}$. Collectively, these tuples constitute the reliability dataset, which provides reliable supervision for training BrainVLM to generate well-calibrated, confidence-aware predictions.

Confidence-aware reliable BrainVLM finetuning

After the construction of the reliability dataset, we then distilled the consensus-derived reliability knowledge into our reliable BrainVLM through confidence-aware reliable fine-tuning. Specifically, as illustrated in Extended Data Fig. 2c, we transformed the vanilla BrainVLM into reliable BrainVLM by parameter-efficient fine-tuning on our constructed reliability dataset. In this process, we also reformulated the output scheme to generate an additional confidence score, yielding responses such as: “Report:..., diagnosis: < tumor_i >. My confidence: 80%”. Crucially, we employed our curated tuples $(x_i, y_{\text{cons}}^{(i)}, s_{\text{con}}^{(i)})$ rather than ground-truth pairs $(x_i, y_{\text{true}}^{(i)})$ as supervision signals, enabling simultaneous learning of diagnostic predictions and their associated confidence levels. During this fine-tuning process, we updated only the projector, LR-HR cross-attention module, and LoRA parameters within the LLaMA model, keeping both the vision encoder and LLaMA backbone frozen. The model was trained for two epochs on the reliability-annotated dataset.

Model inference

Let $\mathcal{S} = \{\text{T1}, \text{T1c}, \text{T2}, \text{FLAIR}\}$ denote the set of MRI sequences. For each sequence $s \in \mathcal{S}$, let $\mathcal{V}_s \subseteq \{\text{axial}, \text{sagittal}, \text{coronal}\}$ represent the available views for s ; $\mathcal{V}_s$ may be empty if the corresponding sequence is missing. Given a patient, the complete input MRI space can be formulated as

$$\mathcal{X} = \{(x_s^v, s, v) \mid s \in \mathcal{S}, v \in \mathcal{V}_s\}, \quad (6)$$

where x_s^v denotes the MRI image corresponding to sequence s and view v . As described in Clinical MRI sequence integration Section), during the inference time, BrainVLM takes five core MRI sequences—T1 (axial or another view), T1c from the same view as T1, T2, FLAIR, and an additional T1c from a different view than T1—along with patient metadata and task-specific instructions as input. Formally, the five core MRI sequences C_m satisfy:

$$C_m = \{(x_{\text{T1}}^{v_0}, \text{T1}, v_0), (x_{\text{T1c}}^{v_0}, \text{T1c}, v_0), (x_{s_3}^{v_3}, s_3, v_3), (x_{s_4}^{v_4}, s_4, v_4), (x_{s_5}^{v_5}, s_5, v_5)\} \quad (7)$$

where $v_0, v_3, v_4, v_5 \in \mathcal{V}_s$ are chosen from available views, and the selection follows these criteria:

- $v_0 \in \mathcal{V}_{\text{T1}}$ is any available view of T1 (e.g., axial, sagittal, or coronal).
- The first two items are T1 and T1c from the same view v_0 .
- For s_3 and s_4 , select from $\{\text{T2}, \text{FLAIR}\}$ according to availability:

- If both T2 and FLAIR are available, then $s_3 = \text{T2}$, $s_4 = \text{FLAIR}$, with $v_3 \in \mathcal{V}_{\text{T2}}$, $v_4 \in \mathcal{V}_{\text{FLAIR}}$.
- If only one of T2 or FLAIR is available, assign it to s_3 and select any other available MRI sequence for s_4 .
- If neither T2 nor FLAIR is available, select any other available MRI sequences for s_3 and s_4 .

- For s_5 , prioritize selecting T1c from any view $v_5 \in \mathcal{V}_{\text{T1c}}$ such that $v_5 \neq v_0$. If not available, select any unused view from the remaining available sequences.

This selection strategy dynamically adapts to the available sequences and views for each patient, ensuring robust performance even in the presence of missing sequences. During inference, BrainVLM takes the selected five core MRI sequences, along with patient metadata and task-specific instructions, as input. It then generates a diagnostic prediction, a radiology report, and an associated confidence score.

Adaptive sequence sampling and aggregation. In real-world clinical practice, MRI studies frequently employ thick-slice acquisitions spanning multiple anatomical planes (e.g., axial, sagittal, and coronal views for both T1 and T2 sequences), yielding studies exceeding the standard five-sequence input. To address this variability, we define an adaptive sequence sampling strategy as follows.

Let $\mathcal{C}$ be the set of all valid five-sequence combinations constructed from $\mathcal{X}$ according to the above selection criteria:

$$\mathcal{C} = \{C_m \subset \mathcal{X} \mid |C_m| = 5 \text{ and } C_m \text{ satisfies the selection constraints above}\}. \quad (8)$$

For each $C_m \in \mathcal{C}$, we define the model inference function:

$$(y_m, r_m, s_{con}^m) = f_\theta(C_m, d, t), \quad (9)$$

where f_θ defines the reliable BrainVLM parameters, d denotes patient metadata, t denotes the task-specific prompt, y_m is the predicted clinical diagnosis, r_m is the generated radiology report, and s_{con}^m is the confidence score. To aggregate results across all sampled combinations, we define the final clinical diagnosis $\hat{y}^{(1)}$ as the most frequent prediction (“majority voting”) among all $\{y_m\}_{C_m \in \mathcal{C}}$:

$$\hat{y}^{(1)} = \text{mode}(\{y_m\}_{m=1}^M) \quad (10)$$

where $\text{mode}(\cdot)$ indicates the statistical mode (most frequent element) in the ensemble. Here $\hat{y}^{(1)}$ is also known as the Top 1 prediction. The final confidence $\hat{s}_{con}^1$ that associated with the predicted diagnosis $\hat{y}^{(1)}$ can be aggregated as:

$$\hat{s}_{con}^{(1)} = \frac{1}{N_1} \sum_{m=1}^M s_{con}^m \cdot \delta(y_m, \hat{y}^{(1)}), \quad N_1 = \sum_{m=1}^M \delta(y_m, \hat{y}^{(1)}), \quad (11)$$

where $\delta(a, b) = 1$ if $a = b$, else 0. The final radiology report is generated by aggregating reports associated with the Top 1 diagnosis. Specifically, since our report are structured into predefined sections (e.g., T1 signal characteristics), we apply majority voting within each section to select the most frequent content among all reports associated with the Top 1 predicted diagnosis $r_m \mid y_m = \hat{y}^{(1)}$.

This adaptive ensemble strategy ensures that the model can robustly process heterogeneous and incomplete clinical MRI acquisitions by (1) systematically sampling all valid five-sequence input sets from the available MRI space $\mathcal{X}$, (2) independently inferring on each combination, and (3) performing robust aggregation via majority voting to determine the final (Top 1) clinical prediction.

Confidence-triggered Top 2 supplementary diagnosis

In clinical workflows, neuroradiologists may consider multiple diagnostic possibilities when interpreting clinically ambiguous cases, such as “considering meningioma or glioma”. To emulate this process, we implement a confidence-triggered supplementary diagnosis mechanism. If the confidence score $\hat{s}_{con}^{(1)}$ for the Top 1 prediction $\hat{y}^{(1)}$ falls below a predefined threshold (set at 75% in this study), a secondary (Top 2) diagnosis is also reported. The Top 2 diagnosis $\hat{y}^{(2)}$ is defined as: $\hat{y}^{(2)} = \text{mode}(\{y_m \mid y_m \neq \hat{y}^{(1)}\})$. Our experimental results in Fig. 1c, Fig. 3c and Fig. 3d in the main manuscript demonstrate the effectiveness of this approach in enhancing diagnostic reliability in ambiguous cases.

Finetuning of molecular subgroup classification task

In this phase, we fine-tuned BrainVLM using the stage-3 checkpoint, leveraging the adult-type diffuse glioma data from BrainTumor46K. We followed the MRI sequences integration strategy defined in the training stage and inference pipeline for molecular classification training and prediction. In this training stage, we updated only the projector, LR-HR cross-attention module and LoRA parameters within the LLaMA model, keeping other parameters frozen. The learning rate was set to 1×10^{-4} , using the AdamW optimizer with a weight decay λ of 0.05.

The results demonstrated BrainVLM’s robust capability to differentiate molecular subtypes of adult diffuse gliomas. BrainVLM achieved a frequency-weighted F_1 score of 0.92 (95% CI: 0.91–0.93, Extended Data Fig. 5c) in the threefold validation on the Xiangya primary dataset with 494 patients, outperforming state-of-the-art models Merlin ($F_1 = 0.76$, 95% CI: 0.75–0.77) and RadFM ($F_1 = 0.74$, 95% CI: 0.73–0.75). Across molecular subgroups, BrainVLM demonstrated superior performance: IDH-w GBM ($F_1 = 0.95$ vs. 0.88–0.91), IDH-m Astro ($F_1 = 0.89$ vs. 0.53–0.62), and IDH-m ODG ($F_1 = 0.84$ vs. 0.58–0.63). In external validation on a dataset of 138 patients, BrainVLM achieved a frequency-weighted F_1 score of 0.88 (Extended Data Fig. 5c), surpassing Merlin ($F_1 = 0.73$) and RadFM ($F_1 = 0.69$). Consistent with primary validation, BrainVLM showed higher performance across subgroups: IDH-w GBM ($F_1 = 0.94$ vs. 0.82–0.85), IDH-m Astro ($F_1 = 0.79$ vs. 0.32–0.44), and IDH-m ODG ($F_1 = 0.69$ vs. 0.27–0.34).

Implementation details

We employed AdamW¹¹⁰ with weight decay regularization ($\lambda = 0.05$) across all training stages, coupled with a cosine decay scheduler and 300-step linear warmup. During the training stage 1 (2D MRI-text representation learning), the learning rate was linearly scaled from 1×10^{-6} to 1×10^{-4} , and the batch size was set to 256. For training stage 2 (hybrid 2D-3D training) and stage 3 (3D volumetric learning), the learning rate followed the same warmup initialization (1×10^{-6}). However, the peak learning rate was set to 3×10^{-5} for training stage 2 and 1×10^{-5} for training stage 3, respectively. We set the batch size to 1 on each GPU device for both training stages 2 and 3, resulting in a total effective batch size of 4. All experiments and the implementation of the BrainVLM were conducted using Python (3.9.1). We also utilized torch (2.0.0), transformers (4.45.2), huggingface-hub (0.29.2), tokenizers (0.20.3) and peft (0.2.0) for model training. For image processing and data augmentation, scikit-image (0.24.0), SimpleITK (2.4.1), monai (1.4.0), numpy (1.26.4), opencv-python (4.7.0) and nibabel (5.3.2) were applied. We used 8 NVIDIA 4090 GPUs for training stage1, 4 NVIDIA L40s GPUs for other training stages.

Competing methods

We compared BrainVLM with three state-of-the-art VLMs: RadFM³¹, Merlin³², and ChatGPT-4o³³. RadFM is a radiology foundation model pretrained on 15.5 million 2D medical images and 500,000 3D medical scans. RadFM utilizes a 12-layer 3D ViT as a vision encoder to process both 2D and 3D data, and uses MedLLaMA-13B as the large language model for reasoning. For our experiments, we fine-tuned the pre-trained RadFM³¹ on our BrainTumor46K dataset, updating the LoRA parameters of MedLLaMA-13B and the projection module. Merlin is a 3D vision-language foundation model developed for abdominal CT interpretation. It was pretrained on both radiology reports and structured electronic health records (EHRs), using a 3D ResNet152 as the vision encoder and RadLlama-7B as the language model³². The vision encoder was pretrained on a large-scale paired dataset of 3D CT scans and text reports using the InfoNCE loss. For our experiments, we fine-tuned Merlin’s vision encoder based on its pre-trained 3D ResNet152 weights, and applied LoRA-based fine-tuning to the RadLlama-7B language model. Both RadFM and Merlin are designed for single-modality radiology inputs. For a fair comparison, we adopted the same input strategy in BrainVLM as in these baseline models. Specifically, we provided five MRI sequences as input, processed each sequence through the respective vision encoders, and concatenated the extracted features before passing them to the language model. To ensure a rigorous and fair comparison, we extensively fine-tuned all baseline models on the BrainTumor46K dataset with sufficient hyperparameter optimization, training each model until convergence.

ChatGPT-4o³³ does not natively support 3D medical imaging input. Therefore, for each test case, experienced clinicians manually selected the most representative tumor-containing slice from each MRI sequence (four to five sequences per case). These clinician-validated slices were combined and provided as a set of 2D images to ChatGPT-4o for report generation and diagnostic prediction. It is important to note that this approach simplifies the diagnostic task, as tumor-containing frames are pre-identified by experts, whereas BrainVLM, RadFM, and Merlin operate directly on the full 3D raw MRI volumes without manual selection.

Quantitative assessment and statistical analysis

We evaluated BrainVLM’s diagnostic capabilities by comparing its outputs with preoperative board-certified neuroradiologists’ interpretations (derived from clinical radiology reports) and multiple state-of-the-art VLMs (RadFM³¹, Merlin³², ChatGPT-4o³³), using postoperative pathological diagnoses as the gold standard. Tumor classification performance was quantified using sensitivity, specificity, precision, Cohen’s $Kappa$, F_1 , and Area Under the Curve (AUC). For report generation evaluation, we used both the common language metrics (Bilingual Evaluation Understudy-4 (BLEU-4⁴⁰)) and clinical relevance metrics

(Ratescore⁴² and RadGraph-XL⁴¹). We applied one-tailed Wilcoxon signed-rank tests with 95% confidence intervals to calculate the statistical difference. Following previous work¹¹¹⁻¹¹⁴, we applied non-parametric bootstrap resampling (1,000 replicates) to estimate 95% CIs, except in AI-augmentation experiments, where CIs were averaged across 12 neuroradiologists.

Given true-positive (TP), false-positive (FP), true-negative (TN) and false-negative (FN) rates, the precision, sensitivity (recall), specificity, and F_1 score of each class are calculated as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (12)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad (13)$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad (14)$$

$$F_1\text{-score} = \frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}, \quad (15)$$

$$\text{Cohen's Kappa} = \frac{P_o - P_e}{1 - P_e}, \quad (16)$$

where,

$$P_o = \frac{TP + TN}{TP + FP + TN + FN} \text{ and } P_e = \frac{(TP + FP) \cdot (TP + FN) + (TN + FN) \cdot (TN + FP)}{(TP + FP + TN + FN)^2}, \quad (17)$$

To assess the BrainVLM's overall multi-label brain tumor classification capability, we computed the class frequency-weighted mean for both the F_1 score and the AUC. More specifically, given the F_{1i} score for class i , the frequency-weighted F_1 score across total C classes are defined as:

$$\text{Weighted } F_1 = \sum_{i=1}^C \frac{n_i}{N} \times F_{1i}. \quad (18)$$

where n_i is the number of patients in class i , N is the total number of patients across all classes.

BLEU-4 is a widely used metric for evaluating lexical similarity between machine-generated and human reference texts.⁴⁰ Unlike traditional natural language metrics, both RadGraph-XL⁴¹ and Ratescore⁴² prioritize domain-specific content critical to radiology, making them particularly suitable for evaluating automated report generation systems. RaTEScore⁴² evaluates clinical reports by identifying medical entities across imaging modalities and regions, and then calculates a clinical relevance-weighted similarity score. Similarly, RadGraph-XL⁴¹ is a metric designed to evaluate the quality of radiology reports by assessing their semantic and clinical accuracy. It constructs a graph-based representation of entities and relationships within a report, capturing anatomical findings, observations, and their interconnections, such as whether a finding is present, absent, or uncertain. By comparing these graphs between generated and reference reports, RadGraph-XL quantifies similarity using an F_1 score, providing a robust measure of how well a generated report conveys clinically relevant information.

Robustness of BrainVLM to missing MRI sequences

BrainVLM performs inference using four standard MRI sequences (T1, T1c, T2, and FLAIR). In clinical practice, incomplete acquisition of MRI sequences (e.g., missing an entire sequence) frequently occurs due to insufficient clinician awareness or contraindications. To evaluate the model's robustness to such scenarios, we conducted an ablation study by systematically excluding one of the four standard sequences. For each excluded sequence, we maintained the required five-image input structure by randomly selecting an additional view from the remaining sequences. For example, when removing T1, we substituted an extra view from T1c, T2, or FLAIR (e.g., supplementary sagittal T1c). For fair comparison, our analysis included only patients meeting the five-image requirement after sequence removal. This resulted in 2,701 patients in the primary test dataset and 870 patients in the external test dataset. As shown in Extended Data Table 5 and Extended Data Fig. 5 (bottom panel), BrainVLM achieves diagnostic performance comparable to the full-sequence baseline in scenarios missing T2, FLAIR, or T1 sequences, with less than a 5% overall reduction in F_1 score. However, the unavailability of T1c sequences caused a 13.4% F_1 -score decline, underscoring its critical role in brain tumor assessment.

Furthermore, we employed Shapley value analysis¹¹⁵ to evaluate the contribution of each input sequence (T1, T1c, T2, and FLAIR) to the model's decisions. As shown in Extended Data Fig. 5a, the analysis revealed consistent patterns with our ablation results: omission of the T1c sequence led to the most substantial decline in diagnostic performance across all tumor types, highlighting its critical importance. Additionally, EMB classification exhibited uniform sensitivity to the removal of any single sequence (approximately 35 % mean performance drop), while GCN and MEN predictions remained stable when T2 or

FLAIR was missing. In contrast, CPT predictions were more dependent on T1, T1c and FLAIR, but showed resilience to the exclusion of T2. These findings align well with clinical diagnostic experience, where specific sequences are known to be more informative for certain tumor types.

Data availability

The online training data are available on their corresponding website. We will publish the scripts for pre-processing the online dataset upon publication of this manuscript. The institutional training data that support the findings of this study are available from the corresponding author upon request, approved by clinical trial number NCT07126821. The detailed publicly available resources can be accessed at: PubMed Central (<https://pmc.ncbi.nlm.nih.gov/>), Ctisus (<https://www.ctisus.com/>), ImageCLEFmedical23 (<https://www.imageclef.org/2023/medical/caption>), Br35H (<https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection>), Figshare Brain Tumor Dataset (<https://www.kaggle.com/datasets/ashkhagan/figshare-brain-tumor-dataset>), Brain Tumor Classification (<https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri>), Brain Tumor MRI8 images 44 classes (<https://www.kaggle.com/datasets/ferando2rad/brain-tumor-mri-images-44c>), LGG-1p19qDeletion (<https://www.cancerimagingarchive.net/collection/lgg-1p19qdeletion/>), BraTS23 (<https://synapse.org/brats>), ReMIND (<https://www.cancerimagingarchive.net/collection/remind/>), RHUH-GBM (<https://www.cancerimagingarchive.net/collection/rhuh-gbm/>), UPENN-GBM (<https://www.cancerimagingarchive.net/collection/upenn-gbm/>), GLIS-RT (<https://www.cancerimagingarchive.net/collection/glis-rt/>), QIN GBM Treatment Response (<https://www.cancerimagingarchive.net/collection/qin-gbm-treatment-response/>), LUMIERE (<https://github.com/ysuter/gbm-data-longitudinal>), OpenNeuro Diffuse Gliomas (<https://openneuro.org/datasets/ds004717/versions/1.0.0>), Brain-Tumor- Progression (<https://www.cancerimagingarchive.net/collection/brain-tumor-progression/>), Radiopaedia (<https://radiopaedia.org/home>), RIDER (<https://www.cancerimagingarchive.net/collection/rider-lung-ct/>), ACRIN-DSC- MR-Brain (<https://www.cancerimagingarchive.net/collection/acrin-dsc-mr-brain/>), ACRIN-FMISO-Brain (<https://www.cancerimagingarchive.net/collection/acrin-fmiso-brain/>), Meningioma-SEG-CLASS (<https://www.cancerimagingarchive.net/collection/meningioma-seg-class/>), Brain metastase MRI dataset (<https://www.nature.com/articles/s41597-023-02123-0>), Brain-TR-GammaKnife, AOMIC- ID1000 (<https://openneuro.org/datasets/ds003097>), AOMIC- PIOP1 (<https://openneuro.org/datasets/ds002785>), IXI (<https://brain-development.org/ixi-dataset/>), OpenNeuro Listening Task (<https://openneuro.org/datasets/ds004285/versions/1.0.0>), OpenNeuro Healthy Adults (<https://openneuro.org/datasets/ds002330/versions/1.1.0/1.1.0>), OpenNeuro Visual Audiovisual Speech (<https://openneuro.org/datasets/ds003717/versions/1.0.1>), OpenNeuro Dynamic Passive Threat (<https://openneuro.org/datasets/>), OpenNeuro UCLA Consortium (<https://openneuro.org/datasets/ds000030/versions/00016>), OpenNeuro NIMH Healthy (<https://openneuro.org/datasets/ds004215>), DLBS dataset (<https://openneuro.org/datasets/ds004856/versions/1.0.0>), QTAB dataset (<https://openneuro.org/datasets/ds004146/versions/1.0.4>). Note that some online resources (e.g., Radiopaedia) cannot be directly accessed for AI research purposes. We have obtained official authorization for non-commercial use in this study.

Code availability

Code of BrainVLM are fully available via Github at <https://github.com/HKU-HealthAI/BrainVLM>.

References

- Price, M. *et al.* Cbtrus statistical report: primary brain and other central nervous system tumors diagnosed in the united states in 2017–2021. *Neuro-oncology* **26**, vi1–vi85 (2024).
- Siegel, R. L., Miller, K. D., Wagle, N. S. & Jemal, A. Cancer statistics, 2023. *CA: a cancer journal for clinicians* **73**, 17–48 (2023).
- Miller, K. D. *et al.* Brain and other central nervous system tumor statistics, 2021. *CA: a cancer journal for clinicians* **71**, 381–406 (2021).
- Louis, D. N. *et al.* The 2021 WHO classification of tumors of the central nervous system: a summary. *Neuro-oncology* **23**, 1231–1251 (2021).

5. Goldbrunner, R. *et al.* EANO guidelines for the diagnosis and treatment of meningiomas. *The Lancet Oncol.* **17**, e383–e391 (2016).
6. Weller, M. *et al.* EANO guidelines on the diagnosis and treatment of diffuse gliomas of adulthood. *Nat. reviews Clin. oncology* **18**, 170–186 (2021).
7. Karschnia, P. *et al.* The oncological role of resection in newly diagnosed diffuse adult-type glioma defined by the who 2021 classification: a review by the RANO resect group. *The Lancet Oncol.* **25**, e404–e419 (2024).
8. Ferreri, A. J. *et al.* Primary central nervous system lymphoma. *Nat. Rev. Dis. Primers* **9**, 29 (2023).
9. Frappaz, D. *et al.* EANO, SNO and Euracan consensus review on the current management and future development of intracranial germ cell tumors in adolescents and young adults. *Neuro-oncology* **24**, 516–527 (2022).
10. Horbinski, C. *et al.* NCCN guidelines® insights: central nervous system cancers, version 2.2022: featured updates to the NCCN guidelines. *J. Natl. Compr. Cancer Netw.* **21**, 12–20 (2023).
11. McIsaac, M., Buchan, J., Abu-Agla, A., Kavar, R. & Campbell, J. Global strategy on human resources for health: workforce 2030—a five-year check-in. *Hum. Resour. for Heal.* **22**, 68 (2024).
12. Zhang, K. *et al.* A generalist vision–language foundation model for diverse biomedical tasks. *Nat. medicine* 1–13 (2024).
13. Wang, J. *et al.* Self-improving generative foundation model for synthetic medical image generation and clinical applications. *Nat. Medicine* **31**, 609–617 (2025).
14. Xiang, J. *et al.* A vision–language foundation model for precision oncology. *Nature* 1–10 (2025).
15. Xue, C. *et al.* AI-based differential diagnosis of dementia etiologies on multimodal data. *Nat. Medicine* **30**, 2977–2989 (2024).
16. Hu, D. *et al.* p95her2, a truncated form of the her2 oncoprotein, drives an immunosuppressive program in her2+ breast cancer that limits trastuzumab deruxtecan efficacy. *Nat. Cancer* 1–21 (2025).
17. Dorfner, F. J., Patel, J. B., Kalpathy-Cramer, J., Gerstner, E. R. & Bridge, C. P. A review of deep learning for brain tumor analysis in MRI. *NPJ Precis. Oncol.* **9**, 2 (2025).
18. Bi, W. L. *et al.* Artificial intelligence in cancer imaging: clinical challenges and applications. *CA: a cancer journal for clinicians* **69**, 127–157 (2019).
19. McKinney, S. M. *et al.* International evaluation of an AI system for breast cancer screening. *Nature* **577**, 89–94 (2020).
20. Mikhael, P. G. *et al.* Sybil: a validated deep learning model to predict future lung cancer risk from a single low-dose chest computed tomography. *J. Clin. Oncol.* **41**, 2191–2200 (2023).
21. Baid, U. & *et al.* The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification. *arXiv preprint arXiv:2107.02314* (2021).
22. LaBella, D. *et al.* Brain tumor segmentation (BraTS) challenge 2024: meningioma radiotherapy planning automated segmentation. *arXiv preprint arXiv:2405.18383* (2024).
23. Menze, B. H. *et al.* The multimodal brain tumor image segmentation benchmark (brats). *IEEE transactions on medical imaging* **34**, 1993–2024 (2014).
24. Moon, H. H. *et al.* Generative AI in glioma: ensuring diversity in training image phenotypes to improve diagnostic performance for idh mutation prediction. *Neuro-oncology* **26**, 1124–1135 (2024).
25. Choi, Y. S. *et al.* Fully automated hybrid approach to predict the idh mutation status of gliomas via deep learning and radiomics. *Neuro-oncology* **23**, 304–313 (2021).
26. Savage, T. *et al.* Large language model uncertainty measurement and calibration for medical diagnosis and treatment. *medRxiv* 2024–06 (2024).
27. Khalighi, S. *et al.* Artificial intelligence in neuro-oncology: advances and challenges in brain tumor diagnosis, prognosis, and precision treatment. *NPJ precision oncology* **8**, 80 (2024).
28. Gu, B., Desai, R. J., Lin, K. J. & Yang, J. Probabilistic medical predictions of large language models. *npj Digit. Medicine* **7**, 367 (2024).
29. Wang, Y.-R. J. *et al.* Advancing presurgical non-invasive molecular subgroup prediction in medulloblastoma using artificial intelligence and MRI signatures. *Cancer Cell* **42**, 1239–1257 (2024).
30. Havaei, M. *et al.* Brain tumor segmentation with deep neural networks. *Med. image analysis* **35**, 18–31 (2017).

31. Wu, C., Zhang, X., Zhang, Y., Wang, Y. & Xie, W. Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data. *Nat. Commun.* **16**, 7866 (2025).
32. Blankemeier, L. *et al.* Merlin: a vision language foundation model for 3D computed tomography. *Res. Sq.* rs-3 (2024).
33. OpenAI. GPT-4. <https://openai.com/research/gpt-4> (2023).
34. Whitfield, B. T. & Huse, J. T. Classification of adult-type diffuse gliomas: Impact of the world health organization 2021 update. *Brain pathology* **32**, e13062 (2022).
35. Wu, J. *et al.* Evolving cell states and oncogenic drivers during the progression of idh-mutant gliomas. *Nat. Cancer* **6**, 145–157 (2025).
36. Zhu, M. *et al.* A three-classification machine learning model for non-invasive prediction of molecular subtypes in diffuse glioma: a two-center study. *Quant. Imaging Medicine Surg.* **15**, 5752 (2025).
37. Rao, V. M. *et al.* Multimodal generative ai for medical image interpretation. *Nature* **639**, 888–896 (2025).
38. Kim, Y. *et al.* Medical hallucinations in foundation models and their impact on healthcare. *arXiv preprint arXiv:2503.05777* (2025).
39. Bai, F., Du, Y., Huang, T. *et al.* M3d: advancing 3d medical image analysis with multi-modal large language models. *arXiv preprint arXiv:2404.00578* (2024).
40. Papineni, K., Roukos, S., Ward, T. & Zhu, W.-J. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, 311–318 (2002).
41. Delbrouck, J.-B. *et al.* RadGraph-XL: a large-scale expert-annotated dataset for entity and relation extraction from radiology reports. In *Findings of the Association for Computational Linguistics ACL 2024*, 12902–12915 (2024).
42. Zhao, W. *et al.* RaTEScore: a metric for radiology report generation. *arXiv preprint arXiv:2406.16845* (2024).
43. Smits, M. MRI biomarkers in neuro-oncology. *Nat. Rev. Neurol.* **17**, 486–500 (2021).
44. Abd-Allah, M. K., Awad, A. I., Khalaf, A. A. & Hamed, H. F. A review on brain tumor diagnosis from MRI images: practical implications, key achievements, and lessons learned. *Magn. resonance imaging* **61**, 300–318 (2019).
45. Vermeulen, C. *et al.* Ultra-fast deep-learned CNS tumour classification during surgery. *Nature* **622**, 842–849 (2023).
46. Hollon, T. & Orringer, D. Using AI to improve the molecular classification of brain tumors (2023).
47. Van den Bent, M. J. *et al.* Primary brain tumours in adults. *The Lancet* **402**, 1564–1579 (2023).
48. Tanno, R. *et al.* Collaboration between clinicians and vision–language models in radiology report generation. *Nat. Medicine* **31**, 599–608 (2025).
49. Rajpurkar, P., Chen, E., Banerjee, O. & Topol, E. J. AI in health and medicine. *Nat. medicine* **28**, 31–38 (2022).
50. Afroogh, S., Akbari, A., Malone, E., Kargar, M. & Alambeigi, H. Trust in AI: progress, challenges, and future directions. *Humanit. Soc. Sci. Commun.* **11**, 1–30 (2024).
51. Abdar, M. *et al.* A review of uncertainty quantification in deep learning: techniques, applications and challenges. *Inf. fusion* **76**, 243–297 (2021).
52. Moor, M. *et al.* Foundation models for generalist medical artificial intelligence. *Nature* **616**, 259–265 (2023).
53. Conte, G. M. *et al.* Generative adversarial networks to synthesize missing T1 and FLAIR MRI sequences for use in a multisequence brain tumor segmentation model. *Radiology* **299**, 313–323 (2021).
54. Grøvik, E. *et al.* Handling missing MRI sequences in deep learning segmentation of brain metastases: a multicenter study. *NPJ digital medicine* **4**, 33 (2021).
55. Rathore, S. *et al.* Brain cancer imaging phenomics toolkit (brain-CaPTk): an interactive platform for quantitative analysis of glioblastoma. In *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: Third International Workshop, BrainLes 2017, Held in Conjunction with MICCAI 2017, Quebec City, QC, Canada, September 14, 2017, Revised Selected Papers 3*, 133–145 (Springer, 2018).
56. Pati, S. *et al.* The cancer imaging phenomics toolkit (CaPTk): technical overview. In *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: 5th International Workshop, BrainLes 2019, Held in Conjunction with MICCAI 2019, Shenzhen, China, October 17, 2019, Revised Selected Papers, Part II 5*, 380–394 (Springer, 2020).
57. Wang, L. *et al.* A lifespan-generalizable skull-stripping model for magnetic resonance images that leverages prior knowledge from brain atlases. *Nat. Biomed. Eng.* 1–16 (2025).

58. Zhu, Z. *et al.* Identifying brain degeneration patterns in early-stage parkinson's disease: a multimodal mri study. *npj Park. Dis.* **11**, 93 (2025).
59. Wang, Y.-R. *et al.* Screening and diagnosis of cardiovascular disease using artificial intelligence-enabled cardiac magnetic resonance imaging. *Nat. Medicine* **30**, 1471–1480 (2024).
60. Ma, J. *et al.* Segment anything in medical images. *Nat. Commun.* **15**, 654 (2024).
61. Joskowicz, L., Cohen, D., Caplan, N. & Sosna, J. Automatic segmentation variability estimation with segmentation priors. *Med. image analysis* **50**, 54–64 (2018).
62. Monje, M. *et al.* Intravenous and intracranial GD2-CAR T cells for H3K27M+ diffuse midline gliomas. *Nature* **637**, 708–715 (2025).
63. Brastianos, P. K. *et al.* BRAF–MEK inhibition in newly diagnosed papillary craniopharyngiomas. *New Engl. J. Medicine* **389**, 118–126 (2023).
64. Adewole, M. *et al.* The brain tumor segmentation (BraTS) challenge 2023: glioma segmentation in sub-Saharan Africa patient population (BraTS-Africa). *arXiv preprint arXiv:2305.19369* (2023).
65. Kazerooni, A. F. *et al.* The brain tumor segmentation (BraTS) challenge 2023: focus on pediatrics (CBTN-CONNECT-DIPGR-ASNR-MICCAI BraTS-PEDs). *arXiv preprint arXiv:2305.17033* (2024).
66. Moawad, A. W. *et al.* The brain tumor segmentation (BraTS-METS) challenge 2023: brain metastasis segmentation on pre-treatment MRI. *arXiv preprint arXiv:2306.00838* (2024).
67. Correia de Verdier, M. *et al.* The 2024 brain tumor segmentation (BraTS) challenge: glioma segmentation on post-treatment MRI. *arXiv preprint arXiv:2405.18368* (2024).
68. Cepeda, S. *et al.* The Río Hortega University Hospital glioblastoma dataset: a comprehensive collection of preoperative, early postoperative and recurrence MRI scans (RHUH-GBM). The Cancer Imaging Archive <https://doi.org/10.7937/4545-c905> (2023).
69. Pedano, N. *et al.* The Cancer Genome Atlas low grade glioma collection (TCGA-LGG) (version 3). The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2016.L4LTD3TK> (2016).
70. Scarpace, L. *et al.* The Cancer Genome Atlas Glioblastoma Multiforme collection (TCGA-GBM) (version 5). The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2016.RNYFYUE9> (2016).
71. Barboriak, D. Data from RIDER NEURO MRI. The Cancer Imaging Archive <http://doi.org/10.7937/K9/TCIA.2015.VOSN3HN1> (2015).
72. Vassantachart, A. *et al.* Segmentation and classification of grade I and II meningiomas from magnetic resonance imaging: an open annotated dataset (Meningioma-SEG-CLASS) (version 1). The Cancer Imaging Archive <https://doi.org/10.7937/0TKV-1A36> (2023).
73. Wang, Y. *et al.* Brain tumor recurrence prediction after gamma knife radiotherapy from MRI and related DICOM-RT: an open annotated dataset and baseline algorithm (Brain-TR-GammaKnife). The Cancer Imaging Archive <https://doi.org/10.7937/xb6d-py67> (2023).
74. Clark, K. *et al.* The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository. *J. digital imaging* **26**, 1045–1057 (2013).
75. Schmainda, K. M. & Prah, M. Data from Brain-Tumor-Progression. The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2018.15quzvnv> (2018).
76. Kinahan, P., Muzi, M., Bialecki, B., Herman, B. & Coombs, L. Data from ACRIN-DSC-MR-Brain. The Cancer Imaging Archive <https://doi.org/10.7937/tcia.2019.zr1pjf4i> (2019).
77. Kinahan, P., Muzi, M., Bialecki, B. & Coombs, L. Data from ACRIN-FMISO-Brain (version 2). The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2018.vohlekok> (2018).
78. Erickson, B., Akkus, Z., Sedlar, J. & Korfiatis, P. Data from LGG-1p19qDeletion (version 2). The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2017.DWEHTZ9V> (2017).
79. Juvekar, P. *et al.* The brain resection multimodal imaging database (ReMIND) (version 1). The Cancer Imaging Archive <https://doi.org/10.7937/3RAG-D070> (2023).
80. Shusharina, N. & Bortfeld, T. Glioma image segmentation for radiotherapy: RT targets, barriers to cancer spread, and organs at risk (GLIS-RT). The Cancer Imaging Archive <https://doi.org/10.7937/TCIA.T905-ZQ20> (2021).

- 1047 **81.** Bakas, S. *et al.* Multi-parametric magnetic resonance imaging (mpmri) scans for de novo glioblastoma (gbm) patients
1048 from the university of pennsylvania health system (upenn-gbm) (version 2). The Cancer Imaging Archive <https://doi.org/10.7937/TCIA.709X-DN49> (2021).
1049
- 1050 **82.** Hamada, A. Br35H: brain tumor detection 2020. Kaggle Dataset [https://www.kaggle.com/datasets/ahmedhamada0/brain](https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection)
1051 [-tumor-detection](https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection) (2020).
- 1052 **83.** Cheng, J. Brain tumor dataset. figshare <https://doi.org/10.6084/m9.figshare.1512427.v8> (2017).
- 1053 **84.** Feltrin, F. Brain tumor MRI images 44 classes. Kaggle Dataset [https://www.kaggle.com/datasets/fernando2rad/brain-tum](https://www.kaggle.com/datasets/fernando2rad/brain-tumor-mri-images-44c)
1054 [or-mri-images-44c](https://www.kaggle.com/datasets/fernando2rad/brain-tumor-mri-images-44c) (2023).
- 1055 **85.** Bhuvaji, S., Kadam, A., Bhumkar, P., Dedge, S. & Kanchan, S. Brain tumor classification (MRI). Kaggle Dataset
1056 <https://www.kaggle.com/dsv/1183165> (2020).
- 1057 **86.** Johns Hopkins University School of Medicine. CTisus. <https://www.ctisus.com/>.
- 1058 **87.** Radiopaedia. Radiopaedia. <https://radiopaedia.org>.
- 1059 **88.** Roberts, R. J. Pubmed central: The genbank of the published literature (2001).
- 1060 **89.** Fan, L. *et al.* The human brainnetome atlas: a new brain atlas based on connectonal architecture. *Cereb. cortex* **26**,
1061 3508–3526 (2016).
- 1062 **90.** Ionescu, B. *et al.* Overview of the imageclef 2021: Multimedia retrieval in medical, nature, internet and social media
1063 applications. In *Experimental IR Meets Multilinguality, Multimodality, and Interaction: 12th International Conference of*
1064 *the CLEF Association, CLEF 2021, Virtual Event, September 21–24, 2021, Proceedings 12*, 345–370 (Springer, 2021).
- 1065 **91.** Ren, S., He, K., Girshick, R. & Sun, J. Faster R-CNN: towards real-time object detection with region proposal networks.
1066 In Cortes, C., Lawrence, N., Lee, D., Sugiyama, M. & Garnett, R. (eds.) *Advances in Neural Information Processing*
1067 *Systems*, vol. 28 (Curran Associates, Inc., 2015).
- 1068 **92.** Subramanian, S. *et al.* MedICaT: a dataset of medical images, captions, and textual references. *arXiv preprint*
1069 *arXiv:2010.06000* (2020).
- 1070 **93.** Smith, R. An overview of the Tesseract OCR engine. In *Ninth international conference on document analysis and*
1071 *recognition (ICDAR 2007)*, vol. 2, 629–633 (IEEE, 2007).
- 1072 **94.** Tong, P. *et al.* Cambrian-1: a fully open, vision-centric exploration of multimodal LLMs. In *Advances in Neural*
1073 *Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024* (2024).
- 1074 **95.** Zhang, S. *et al.* Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific
1075 image-text pairs. *arXiv preprint arXiv:2303.00915* (2023).
- 1076 **96.** Radford, A. *et al.* Learning transferable visual models from natural language supervision. In *International conference on*
1077 *machine learning*, 8748–8763 (PmLR, 2021).
- 1078 **97.** Guo, Z. *et al.* LLaVA-UHD: An LMM perceiving any aspect ratio and high-resolution images. In *Computer Vision -*
1079 *ECCV 2024 - 18th European Conference*, vol. 15141, 390–406 (2024).
- 1080 **98.** Grattafiori, A. *et al.* The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783* (2024).
- 1081 **99.** Wang, W.-Y., Wang, Z., Suzuki, H. & Kobayashi, Y. Seeing is understanding: unlocking causal attention into modality-
1082 mutual attention for multimodal LLMs. *arXiv preprint arXiv:2503.02597* (2025).
- 1083 **100.** Zhou, C. *et al.* Transfusion: Predict the next token and diffuse images with one multi-modal model. *arXiv preprint*
1084 *arXiv:2408.11039* (2024).
- 1085 **101.** Xie, J. *et al.* Show-o: One single transformer to unify multimodal understanding and generation. *arXiv preprint*
1086 *arXiv:2408.12528* (2024).
- 1087 **102.** Lowekamp, B. C., Chen, D. T., Ibáñez, L. & Blezek, D. The design of SimpleITK. *Front. neuroinformatics* **7**, 45 (2013).
- 1088 **103.** Brinjikji, W. *et al.* MRI findings of disc degeneration are more prevalent in adults with low back pain than in asymptomatic
1089 controls: a systematic review and meta-analysis. *Am. J. Neuroradiol.* **36**, 2394–2399 (2015).
- 1090 **104.** Lambin, P. *et al.* Radiomics: the bridge between medical imaging and personalized medicine. *Nat. reviews Clin. oncology*
1091 **14**, 749–762 (2017).
- 1092 **105.** Hu, E. J. *et al.* Lora: low-rank adaptation of large language models (2021). [2106.09685](https://arxiv.org/abs/2106.09685).
- 1093 **106.** Despotović, I., Goossens, B. & Philips, W. MRI segmentation of the human brain: challenges, methods, and applications.
1094 *Comput. mathematical methods medicine* **2015**, 450341 (2015).

- 1095 **107.** Li, C., Wong, C., Zhang, S. *et al.* LLaVA-Med: training a large language-and-vision assistant for biomedicine in one day.
1096 arxiv. *arXiv preprint arXiv:2306.00890* (2023).
- 1097 **108.** Amini, A., Vieira, T., Ash, E. & Cotterell, R. Variational best-of-n alignment. *arXiv preprint arXiv:2407.06057* (2024).
- 1098 **109.** Wan, G., Wu, Y., Chen, J. & Li, S. Reasoning aware self-consistency: Leveraging reasoning paths for efficient llm
1099 sampling. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for*
1100 *Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 3613–3635 (2025).
- 1101 **110.** Loshchilov, I. & Hutter, F. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).
- 1102 **111.** Rakocz, N. *et al.* Automated identification of clinical features from sparsely annotated 3-dimensional medical imaging.
1103 *NPJ digital medicine* **4**, 44 (2021).
- 1104 **112.** Tiu, E. *et al.* Expert-level detection of pathologies from unannotated chest x-ray images via self-supervised learning. *Nat.*
1105 *biomedical engineering* **6**, 1399–1406 (2022).
- 1106 **113.** Kehl, K. L. *et al.* Shareable artificial intelligence to extract cancer outcomes from electronic health records for precision
1107 oncology research. *Nat. Commun.* **15**, 9787 (2024).
- 1108 **114.** Ström, P. *et al.* Artificial intelligence for diagnosis and grading of prostate cancer in biopsies: a population-based,
1109 diagnostic study. *The Lancet Oncol.* **21**, 222–232 (2020).
- 1110 **115.** Lundberg, S. M. & Lee, S.-I. A unified approach to interpreting model predictions. *Adv. neural information processing*
1111 *systems* **30** (2017).
- 1112 **116.** Qiu, Z. *et al.* Learning co-plane attention across MRI sequences for diagnosing twelve types of knee abnormalities. *Nat.*
1113 *Commun.* **15**, 7637 (2024).
- 1114 **117.** Suter, Y., Knecht, U., Valenzuela, W. *et al.* The lumiere dataset: longitudinal glioblastoma mri with expert rano evaluation.
1115 *Sci Data* **9**, 768, DOI: [10.1038/s41597-022-01881-7](https://doi.org/10.1038/s41597-022-01881-7) (2022).
- 1116 **118.** Filimonova, E., Pashkov, A., Borisov, N., Kalinovskiy, A. & Rzaev, J. Utilizing the amide proton transfer technique to
1117 characterize diffuse gliomas based on the who 2021 classification of cns tumors. *The Neuroradiol. J.* **37**, 490–499 (2024).
- 1118 **119.** Zolotova, S. V. *et al.* Burdenko's glioblastoma progression dataset (burdenko-gbm-progression) (version 1). The Cancer
1119 Imaging Archive (2023). <https://doi.org/10.7937/EIQP-D183>.
- 1120 **120.** Boxerman, J. L. *et al.* Early post-bevacizumab progression on contrast-enhanced mri as a prognostic marker for overall
1121 survival in recurrent glioblastoma: results from the acrin 6677/rtog 0625 central reader study. *Neuro-oncology* **15**,
1122 945–954 (2013).
- 1123 **121.** Gerstner, E. R. *et al.* Acrin 6684: assessment of tumor hypoxia in newly diagnosed glioblastoma using 18f-fmiso pet and
1124 mri. *Clin. Cancer Res.* **22**, 5079–5086 (2016).
- 1125 **122.** Ocaña-Tienda, B. *et al.* A comprehensive dataset of annotated brain metastasis mr images with clinical and radiomic data.
1126 *Sci. data* **10**, 208 (2023).
- 1127 **123.** Snoek, L. *et al.* "aomic-id1000", DOI: [10.18112/openneuro.ds003097.v1.2.1](https://doi.org/10.18112/openneuro.ds003097.v1.2.1) (2021).
- 1128 **124.** Snoek, L. *et al.* "aomic-piop1", DOI: [10.18112/openneuro.ds002785.v2.0.0](https://doi.org/10.18112/openneuro.ds002785.v2.0.0) (2020).
- 1129 **125.** Silva, S., Lorenzi, M. & Ayed, S.-S. Ixi sample dataset. Mendeley Data, V3, DOI: [10.17632/7kd5wj7v7p.3](https://doi.org/10.17632/7kd5wj7v7p.3) (2022).
- 1130 **126.** Rogers, C. S., Jones, M. S., McConkey, S. & Peelle, J. E. "listening task", DOI: [doi:10.18112/openneuro.ds004285.v1.0.0](https://doi.org/10.18112/openneuro.ds004285.v1.0.0)
1131 (2022).
- 1132 **127.** Sunavsky, A. & Poppenk, J. "neuroimaging predictors of creativity in healthy adults", DOI: [10.18112/openneuro.ds002330.v1.1.0](https://doi.org/10.18112/openneuro.ds002330.v1.1.0) (2020).
- 1133 **128.** JE, P. *et al.* "visual and audiovisual speech perception associated with increased functional connectivity between sensory
1134 and motor regions", DOI: [doi:10.18112/openneuro.ds003717.v1.0.1](https://doi.org/10.18112/openneuro.ds003717.v1.0.1) (2022).
- 1135 **129.** Bilder, R. *et al.* "ucla consortium for neuropsychiatric phenomics la5c study" (2018).
- 1136 **130.** Nugent, A. C. *et al.* "the nimh healthy research volunteer dataset", DOI: [doi:10.18112/openneuro.ds005752.v2.1.0](https://doi.org/10.18112/openneuro.ds005752.v2.1.0) (2025).
- 1137 **131.** Park, D. *et al.* "the dallas lifespan brain study", DOI: [doi:10.18112/openneuro.ds004856.v1.0.0](https://doi.org/10.18112/openneuro.ds004856.v1.0.0) (2024).
- 1138 **132.** Strike, L. T. *et al.* "queensland twin adolescent brain (qtab)", DOI: [doi:10.18112/openneuro.ds004146.v1.0.4](https://doi.org/10.18112/openneuro.ds004146.v1.0.4) (2022).
- 1139 **133.** Zepeda Castellanos J, C. p. p. Case study, radiopaedia.org (accessed on 18 jun 2025). <https://doi.org/10.53347/rID-208152>.
1140

1141 **Acknowledgements**

1142 This study was supported by the National Natural Science Foundation of China (62306253), the Early Career Fund (27207025,
1143 27204623), the Clinical Research Fund of the National Clinical Research Center for Geriatric Disorders (2023LNJJ19), the
1144 Natural Science Foundation of Hunan Province (2023JJ30927), and the Guangdong Natural Science Fund-General Program
1145 (2024A1515010233).

1146 **Author contributions statement**

1147 L.Q. and X.G. conceptualized the study. J.C., Y.W., H.J., L.Q., and A.H. conducted data collection and preprocessing from
1148 public repositories. X.G., Z.C., S.K., B.Z., B.W., Z.X., H.Y., Z.W., W.W., B.T., W.H., W.D., W.Z., M.L., A.H., B.L., L.W.,
1149 and W-w.W. collected data from medical centers. X.G., Z.C., S.K., J.C., Y.W., and H.J. performed data preprocessing for
1150 institutional data. Y.W., J.C., S.Z., H.J., and L.Q. developed and trained the deep learning algorithms and conducted model
1151 analysis. X.G., L.Q., M.I., Y-R.W., F.W., Y.Z., L.Z., and Z.L. provided domain knowledge and interpreted the findings.
1152 L.Q., X.G., Y.W., and J.C. drafted the primary manuscript and prepared the figures, with input from all authors. All authors
1153 contributed to the general discussion, manuscript revision, and approved the final version. L.Q. and X.G. co-supervised the
1154 study.

1155 **Competing interests**

1156 The authors declare no competing interests.

Table 1 | Characteristics of our online dataset, primary dataset, and external test dataset for each brain tumor type.

	Online dataset			Primary dataset				External test dataset				Entire dataset
	No. of subjects	Dimension		No. of subjects	Sex		Age in years Mean $\pm$ Std (range)	No. of subjects	Sex		Age in years Mean $\pm$ Std (range)	
		2D	3D		Male	Female			Male	Female		
Total	20,929	13,096	7,833	8,733	4,198	4,535	45 $\pm$ 19 (1-86)	1,329	662	667	52 $\pm$ 15 (1-86)	30,991
MET	1,554	879	675	735	435	300	56 $\pm$ 11 (7-79)	116	69	47	60 $\pm$ 11 (34-86)	2,405
GCT	81	62	19	332	242	90	19 $\pm$ 13 (1-67)	18	11	7	20 $\pm$ 13 (1-40)	431
GGN	9,161	4,743	4,418	3,028	1,681	1,347	41 $\pm$ 19 (1-86)	414	244	170	50 $\pm$ 15 (5-79)	12,603
MEN	5,363	3,804	1,559	2,395	723	1672	53 $\pm$ 12 (1-81)	355	115	240	56 $\pm$ 13 (8-79)	8,113
TSR	674	614	60	796	402	394	46 $\pm$ 16 (2-81)	186	98	88	51 $\pm$ 15 (5-75)	1,656
MNM	912	457	455	447	234	213	39 $\pm$ 18 (1-78)	76	40	36	43 $\pm$ 16 (8-76)	1,435
CPN	1,675	1,392	283	416	138	278	46 $\pm$ 15 (1-75)	90	43	47	54 $\pm$ 10 (34-75)	2,181
CPT	259	216	43	109	56	53	27 $\pm$ 20 (1-66)	6	1	5	45 $\pm$ 10 (39-62)	374
HEM	535	396	139	164	100	64	47 $\pm$ 22 (2-77)	44	27	17	62 $\pm$ 11 (30-82)	743
EMB	288	165	123	205	131	74	18 $\pm$ 15 (1-70)	18	12	6	18 $\pm$ 15 (3-40)	511
PIN	346	312	34	64	33	31	37 $\pm$ 19 (1-73)	6	2	4	39 $\pm$ 13 (21-57)	416
MEL	81	56	25	42	23	19	49 $\pm$ 14 (14-73)	-	-	-	-	123

In this table, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).

Table 2 | Comparative diagnostic performance across neuroradiologists-only, BrainVLM, and AI-augmented neuroradiologists paradigms. Twelve neuroradiologists with varying years of experience (5 juniors [3-5 years], 4 seniors [5-10 years], 3 experts [10+ years]) participated in this blinded crossover study. Boldface indicates the highest performance metric within each tumor type across AI, neuroradiologists, and AI-augmented neuroradiologists cohorts.

	F_1 score						
	AI model	Junior	Junior-AI	Senior	Senior-AI	Expert	Expert-AI
MET	0.76	0.39	0.64	0.46	0.64	0.88	0.90
GCT	0.71	0.14	0.64	0.40	0.62	0.67	0.77
GGN	0.84	0.63	0.70	0.65	0.75	0.82	0.87
MEN	0.85	0.72	0.83	0.79	0.88	0.92	0.92
TSR	0.89	0.81	0.84	0.79	0.83	0.84	0.87
MNM	0.50	0.0	0.32	0.09	0.33	0.40	0.42
CPN	0.83	0.60	0.75	0.57	0.77	0.64	0.83
CPT	0.67	0.21	0.38	0.22	0.49	0.53	0.67
HEM	0.71	0.16	0.52	0.26	0.66	0.76	0.87
EMB	0.80	0.14	0.67	0.16	0.63	0.47	0.67
PIN	0.40	0.35	0.40	0.37	0.50	0.50	0.50
MEL	0.4	0.0	0.22	0.0	0.32	0.53	0.57
Frequency-weighted F_1	0.77	0.49	0.68	0.52	0.71	0.77	0.84
Accuracy	0.79	0.53	0.71	0.57	0.74	0.79	0.86
Time cost (in total)	23 min	189 min	109 min	161 min	104 min	100 min	75 min

In this table, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).

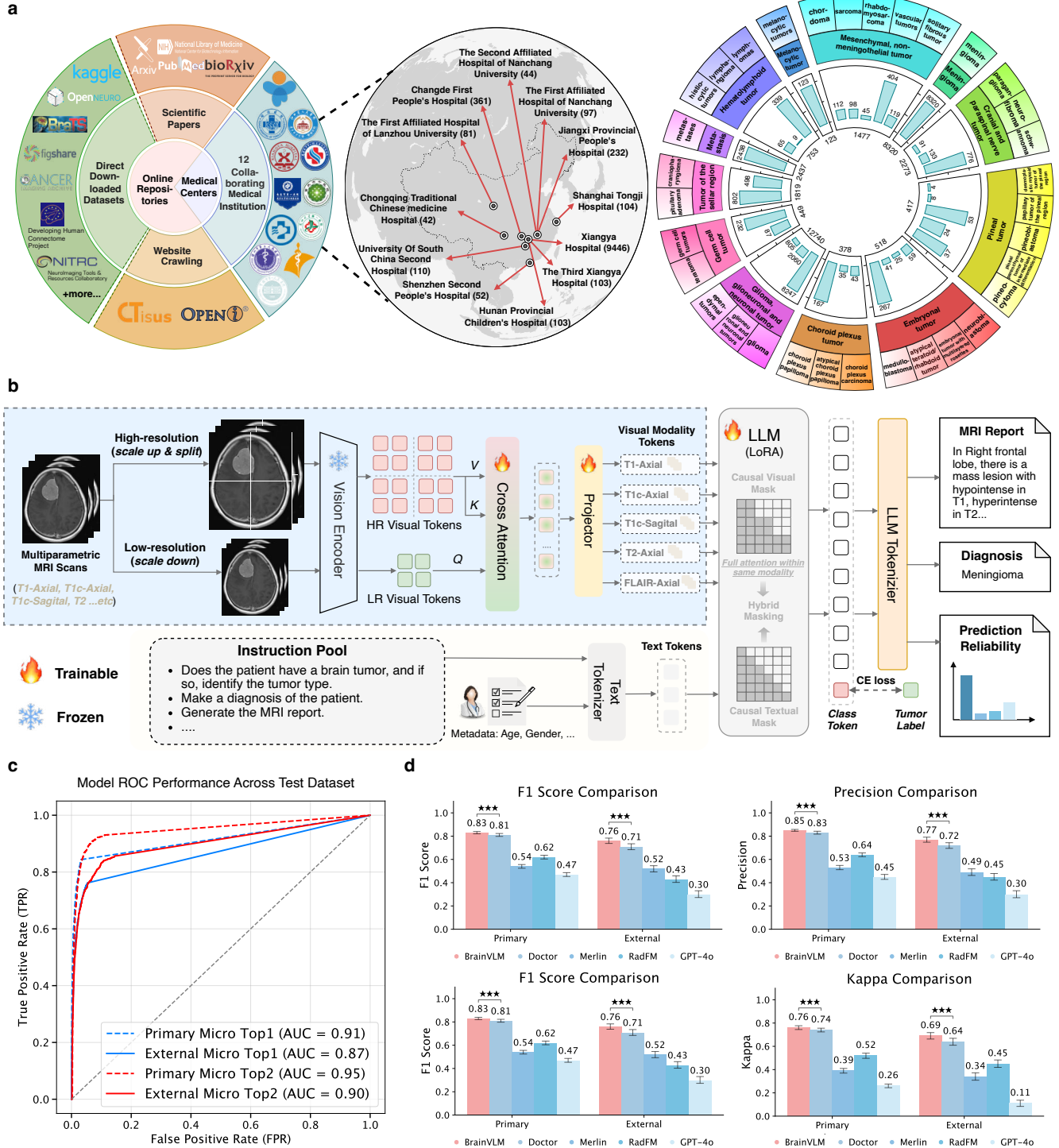


Fig. 1 | Overview of this study. **a**, A multi-modal brain tumor dataset was constructed from 36 online sources and 12 collaborating medical institutions, comprising 46,512 patients. Each case includes either 2D MRI slices or multi-sequence 3D MRI scans (T1-weighted [T1], T1 contrast-enhanced [T1c], T2-weighted [T2], T2-FLAIR [FLAIR]), paired with pathologically confirmed diagnoses, *optional* radiology reports (or text descriptions). The detailed number of patients across 12 brain tumor types and their subtypes is shown on the right panel. **b**, We developed BrainVLM, a vision-language foundation model designed for brain tumor analysis. The model processes multi-parametric MRI scans at different resolutions through a shared vision encoder and integrates text data via an LLM-based architecture. BrainVLM was trained to classify tumor types with quantified diagnostic uncertainty and a radiology report. **c**, ROC curves for Top 1 and Top 2 predictions on the primary and external datasets, with micro-averaged AUCs of 0.91/0.95 and 0.87/0.91, respectively. **d**, BrainVLM outperformed neuroradiologists and baseline models^{31–33} in sensitivity, precision, F_1 score, and Cohen's kappa across both datasets. One-tailed Wilcoxon signed-rank tests were performed to compare BrainVLM with the best-performing alternative model or neuroradiologists. Statistical significance is indicated as ns (not significant, $P > 0.05$), ★ ($P < 0.05$), ★★ ($P < 0.01$), and ★★★ ($P < 0.001$). Bootstrap resampling (1,000 replicates) was used to estimate 95% confidence intervals, shown as error bars around the mean.^{111–113}

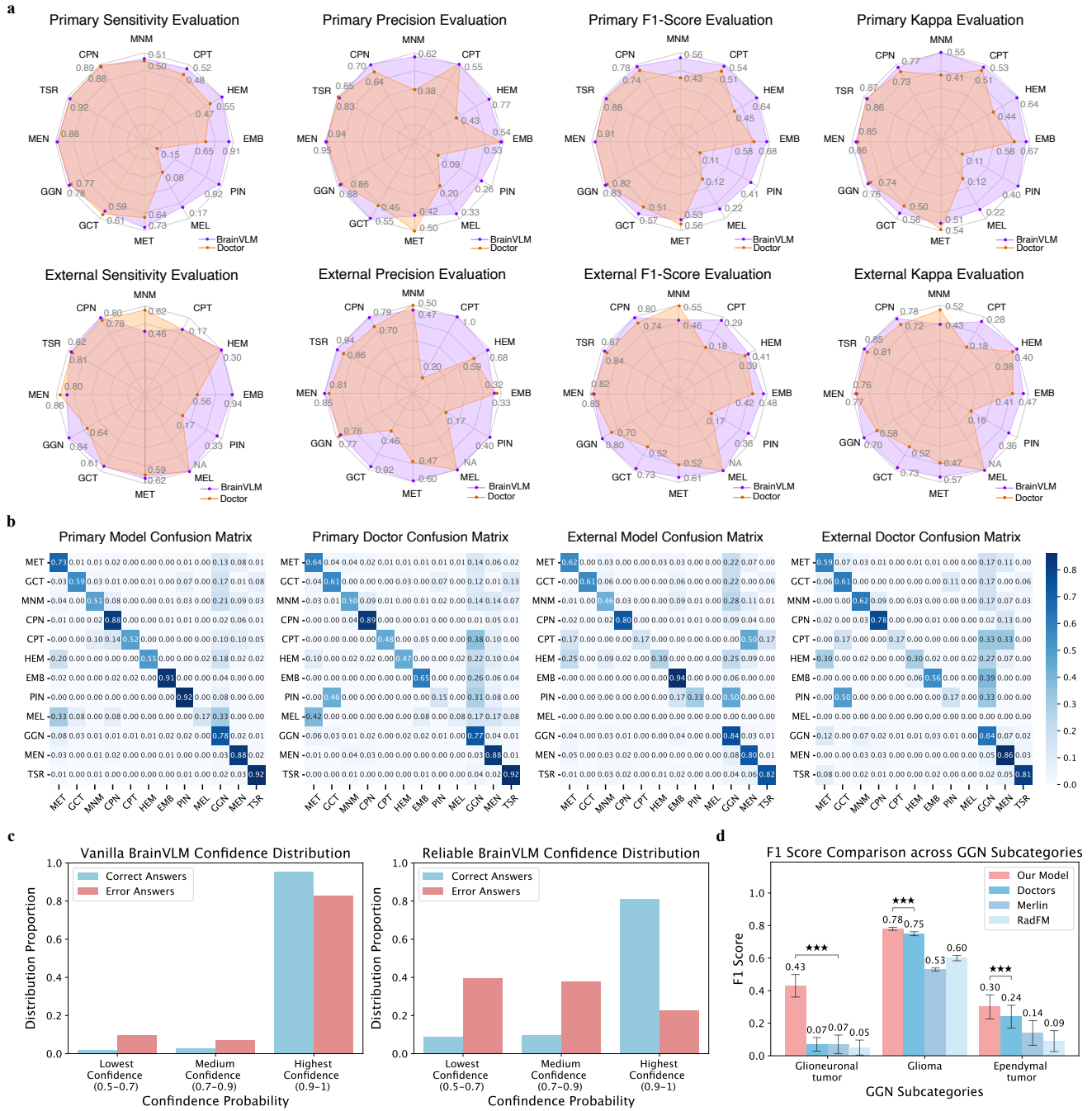


Fig. 2 | Multi-dimensional evaluation of BrainVLM's diagnostic performance. **a**, Performance comparison of BrainVLM and board-certified neuroradiologists on 12 tumor types across primary and external datasets. Model and human expert results are visualized as a dodecagon, with each vertex corresponding to a specific tumor type. For both datasets, we report sensitivity, precision, F_1 score, and Cohen's kappa for each class. BrainVLM demonstrates either superior or comparable performance relative to neuroradiologists on both primary and external cohorts, including for rare tumor types (e.g., EMB, GCT, HEM). **b**, Confusion matrices comparing BrainVLM's predictions to those of neuroradiologists across 12 tumor types in both primary and external datasets. The foundation model showed stronger diagonal concentration than neuroradiologists, indicating improved classification accuracy. Each matrix was normalized by true labels to highlight per-class performance. Notably, the model's diagnostic error patterns showed substantial concordance with those of human experts. **c**, The comparison of confidence score distributions between Vanilla BrainVLM, which does not incorporate our consensus-driven strategy for uncertainty quantification, and Reliable BrainVLM, which benefits from the integration of this strategy. **d**, F_1 score comparison between our model, RadFM³¹, Merlin³², and neuroradiologists in diagnosing the GGN subcategories, which include gliomas, glioblastomas and neuronal tumors, and ependymal tumors. In this figure, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioblastomas, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinous nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).

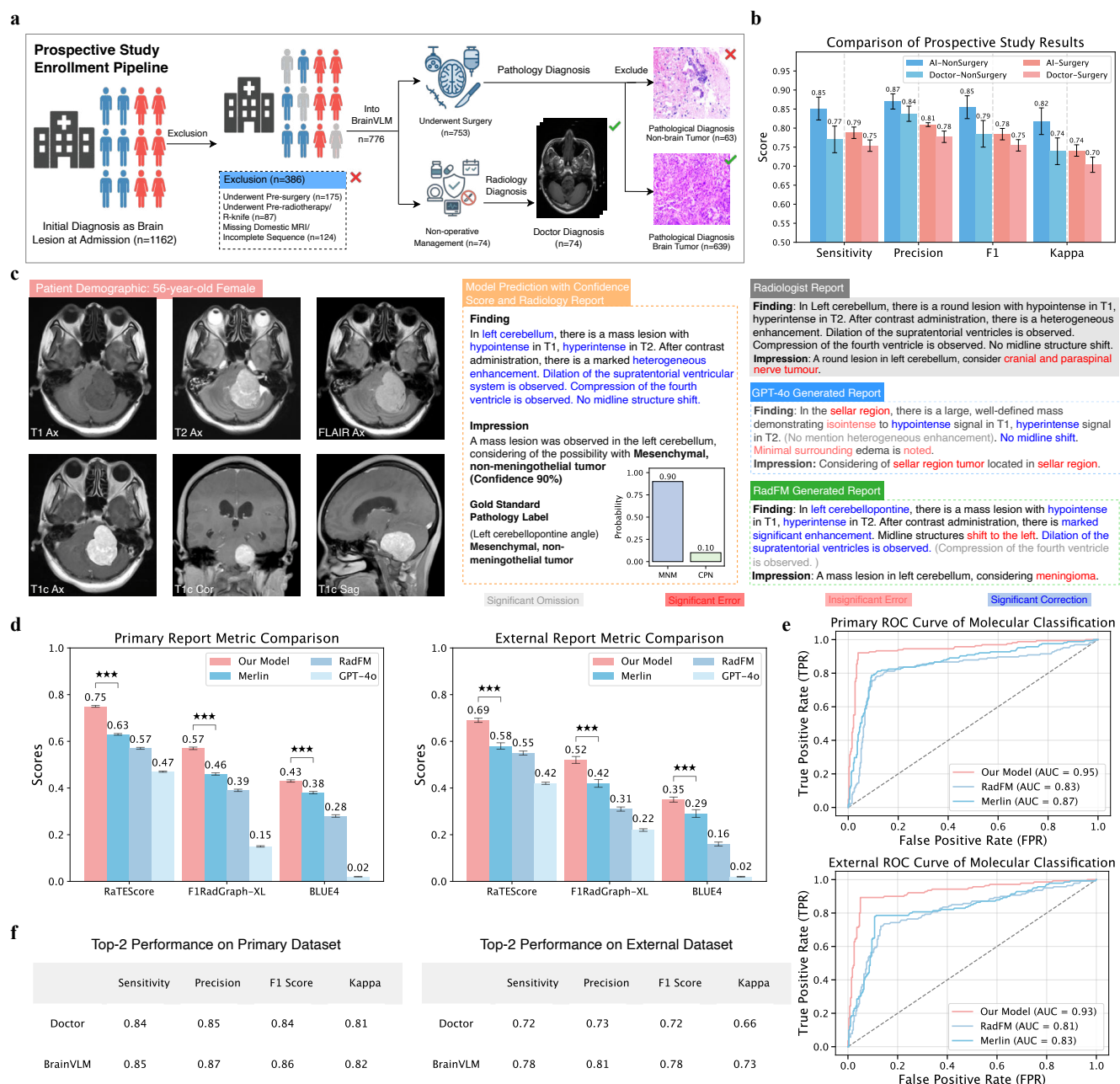


Fig. 3 | Prospective study design and performance comparison in the prospective cohort, as well as comparative evaluation of radiology report generation, molecular subgroup prediction, and top-2 diagnosis prediction across BrainVLM, neuroradiologists, and other AI models in retrospective studies. **a**, Flowchart of prospective patient enrollment. A total of 1,162 patients with suspected brain lesions were prospectively enrolled. After excluding those who had received prior treatment or lacked diagnostic sequences, 831 patients were eligible for evaluation. Among them, 753 underwent surgery; after excluding 63 cases with non-brain tumour pathology, 639 patients with confirmed brain tumours were included for model validation. The remaining 74 patients who received non-operative management were also included for evaluation. **b**, The diagnostic performance of neuroradiologists and our BrainVLM model was compared across both surgery and non-surgery groups. Our BrainVLM demonstrated superior performance in sensitivity, precision, F1 score, and Cohen's kappa. **c**, Comparison of generated reports from ChatGPT-4o³³, RadFM³¹, Merlin³², and our model against human doctor-generated reference reports, highlighting corrections, errors, and omissions. For each of BrainVLM's diagnosis, the associated confidence score is also presented. **d**, Report metric comparison of four models using RaTEScore⁴², F1Radgraph-XL⁴¹, and BLEU4⁴⁰ on both primary and external datasets. **e**, ROC curves comparison for preoperative molecular subgroup prediction in adult-type diffuse gliomas among RadFM³¹, Merlin³², and our model on both primary and external datasets. **f**, Top 2 diagnosis metrics including F_1 score, sensitivity, precision, and Cohen's kappa comparison between our BrainVLM and neuroradiologists.

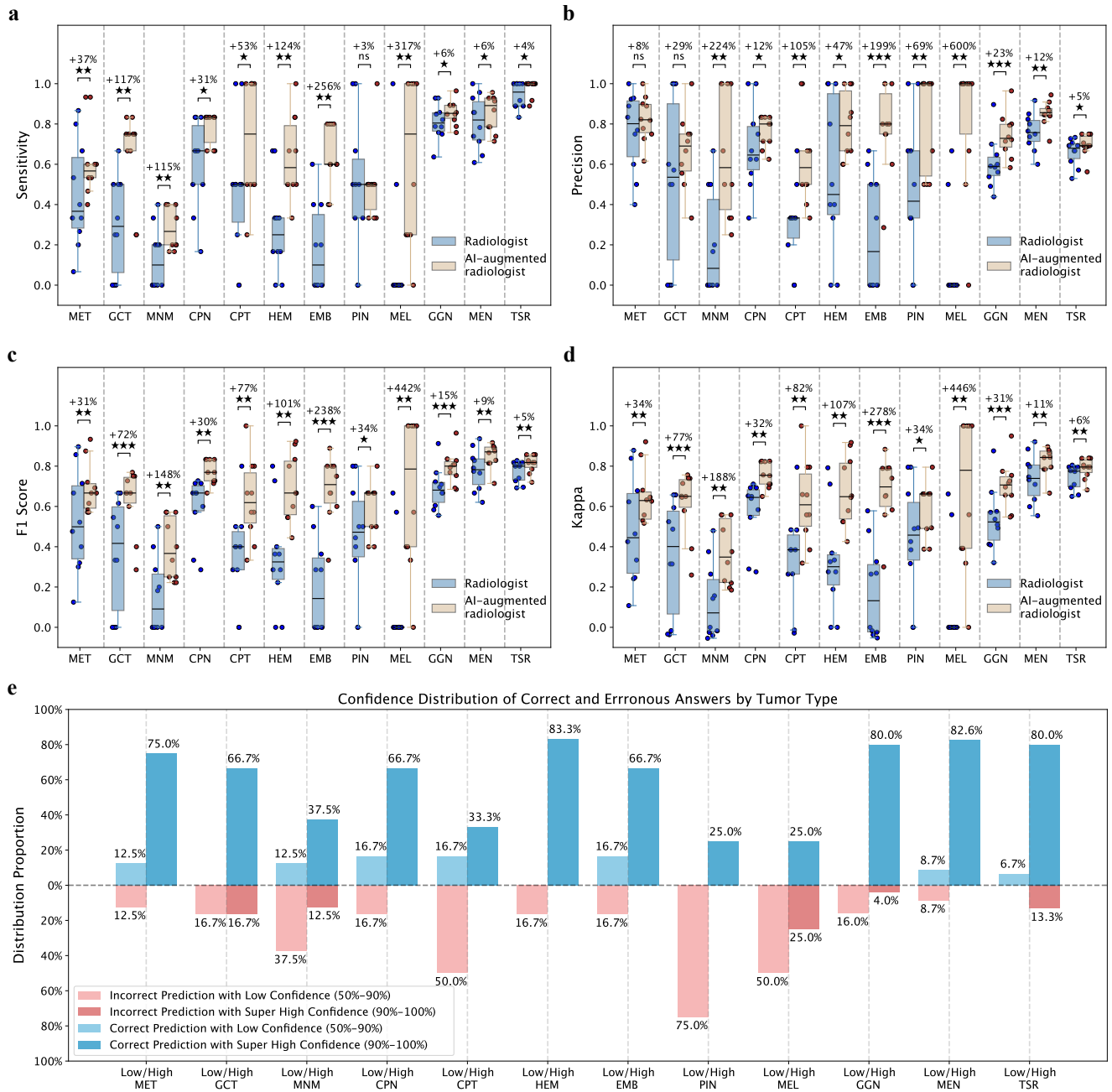


Fig. 4 | Blinded multi-reader study on 123 patients with interpretations by 12 neuroradiologists (5 juniors [3-5 years], 4 seniors [5-10 years], 3 experts [10+ years]) with and without AI assistance. **a**, Confidence score distribution for BrainVLM's predictions on these 123 cases across 12 major WHO CNS5 brain tumor types. Confidence scores were categorized into four groups: correct prediction with super high confidence (0.9-1.0, **dark blue**), correct prediction with low confidence (0.5-0.9, **light blue**), incorrect prediction with super high confidence (0.9-1.0, **dark pink**) and incorrect prediction with low confidence (0.5-0.9, **light pink**). The figure is vertically divided: upper blue bars show confidence distributions for correct predictions, lower pink bars show incorrect ones. Across all cases, BrainVLM assigns high confidence more often to correct predictions (dark blue > light blue), while low confidence is more common among incorrect predictions (light pink > dark pink). Moreover, for tumor types with higher BrainVLM performance (F1 score > 0.65), such as GGN and EMB, the ratio of correct predictions with high confidence reaches 80%, and 67%, respectively. Conversely, for more challenging tumor types (e.g., PIN, MEL), a larger proportion of incorrect predictions are associated with low confidence scores. **b-e**, Comparison between the diagnostic performance of neuroradiologists and AI-augmented neuroradiologists. Box plots summarize sensitivity, precision, F_1 score, and Cohen's kappa for individual neuroradiologists and their AI-augmented counterparts. Pairwise comparisons between each neuroradiologist and their corresponding AI-augmented counterpart were performed using one-tailed Wilcoxon signed-rank tests without correction for multiple comparisons. Statistical significance is indicated as ns (not significant, $P > 0.05$), ★ ($P < 0.05$), ★★ ($P < 0.01$), and ★★★ ($P < 0.001$). In this figure, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinous nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).

Extended Data

Extended Data Table 1 | Characteristics of the primary dataset (Xiangya Hospital) and external test datasets from 11 independent medical centers.

	No. of subjects	Sex		Age in years (range)
		Male	Female	
Total	10,062	4,860 (48%)	5,202 (52%)	53±16 (1–86)
Train dataset	4,936	2,571 (52%)	2,365 (48%)	43±19 (1–86)
Test dataset	5,126	2,289 (45%)	2,837 (55%)	48±17 (1–86)
Xiangya Hospital	8,733	4,198 (48%)	4,535 (52%)	45±19 (1–86)
Changde First People's Hospital	361	193 (53%)	168 (47%)	55±13 (6–86)
Shanghai Tongji Hospital	104	54 (52%)	50 (48%)	54±16 (5–79)
University of South China Second Hospital	110	57 (52%)	53 (48%)	52±15 (6–80)
Shenzhen Second People's Hospital	52	30 (58%)	22 (42%)	49±15 (8–75)
The Third Xiangya Hospital	103	49 (48%)	54 (52%)	52±17 (7–76)
Jiangxi Provincial People's Hospital	232	107 (46%)	125 (54%)	53±15 (13–82)
Chongqing Traditional Chinese medicine Hospital	42	17 (40%)	25 (60%)	58±12 (26–78)
The First Affiliated Hospital of Nanchang University	97	40 (41%)	57 (59%)	46±17 (20–67)
Hunan Provincial Children's Hospital	103	56 (54%)	47 (46%)	28±17 (1–60)
The First Affiliated Hospital of Lanzhou University	81	31 (38%)	50 (62%)	52±13 (8–76)
The Second Affiliated Hospital of Nanchang University	44	28 (64%)	16 (36%)	55±11 (13–74)

In this table, we reported the age of patients using mean ± std.

Extended Data Table 2 | Summary of publicly available brain MRI datasets. The table summarizes a range of datasets, detailing the number of subjects, 2D MRI slices, data modalities (2D/3D), and pathological labels (tumor or healthy). For each dataset, original source information and the specific tasks organized in this study are also provided. Here, “3D Healthy” refers to datasets containing healthy individuals with 3D MRI volumes, while “3D Tumor” refers to datasets containing brain tumor patients with 3D MRI volumes. The PubMed Central dataset includes both normal and brain tumor subjects. For ReMIND dataset, we utilized 108 brain tumor cases, excluding 6 brain disease cases. Please note that, due to policy restrictions (e.g., Radiopaedia), certain portions of the training dataset cannot be shared.

Data Type	Data Resource	No. of subjects	No. of images	Original Information	Organized Tasks
2D web crawling	PubMed Central ⁸⁸	23,849	116,296	Description	2D MRI slice-text description Tumor classification
2D web crawling	Ctiscus ⁸⁶	150	1,124	Description, Diagnosis	2D MRI slice-text description Tumor classification
2D web crawling	ImageCLEFmedical23 ⁹⁰	484	4,631	Description	2D MRI slice-text description Tumor classification
2D Kaggle	Br35H ⁸²	-	2,137	Diagnosis Segment	2D MRI slice-text description Tumor classification
2D Kaggle	Figshare Brain ⁸³ Tumor Dataset	233	3,064	Diagnosis Segment	2D MRI-report Tumor classification
2D Kaggle	Brain Tumor ⁸⁵ Classification	-	3,264	Diagnosis	Tumor classification
2D Kaggle	Brain Tumor MRI ⁸⁴ images 44 classes	-	4,479	Diagnosis	Tumor classification
3D Tumor	LGG-1p19qDeletion ⁷⁸	159	32,829	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	BraTS23 ^{21,22,64-67}	3,263	2,156,589	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	ReMIND ^{79*}	108	60,963	Diagnosis metadata	3D MRI-report Tumor classification
3D Tumor	RHUH-GBM ⁶⁸	40	122,835	Diagnosis	Tumor classification
3D Tumor	UPENN-GBM ⁸¹	630	622,112	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	GLIS-RT ⁸⁰	230	83,196	Diagnosis	Tumor classification
3D Tumor	QIN GBM Treatment Response ¹¹⁶	54	66,988	Diagnosis	Tumor classification
3D Tumor	LUMIERE ¹¹⁷	91	460,155	Diagnosis	Tumor classification
3D Tumor	OpenNeuro (Diffuse Gliomas) ¹¹⁸	42	22,412	Diagnosis	Tumor classification
3D Tumor	Brain-Tumor- Progression ⁷⁴	20	46,573	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	Radiopaedia ⁸⁷	2606	376, 374	Diagnosis Report	3D MRI-report Tumor classification
3D Tumor	Burdenko ¹¹⁹	180	460,524	Diagnosis	Tumor classification
3D Tumor	RIDER ⁷¹	19	115,100	Diagnosis	Tumor classification
3D Tumor	ACRIN-DSC- MR-Brain ¹²⁰	123	1,385,512	Diagnosis	Tumor classification
3D Tumor	ACRIN-FMISO- Brain ¹²⁰	50	508,500	Diagnosis	Tumor classification
3D Tumor	Meningioma- SEG-CLASS ¹²¹	96	83,556	Diagnosis metadata	3D MRI-report Tumor classification

Extended Data Table 2 | Continuation of Table 2. Summary of publicly available brain MRI datasets. The table summarizes a range of datasets, detailing the number of subjects, 2D MRI slices, data modalities (2D/3D), and pathological labels (tumor or healthy). For each dataset, original source information and the specific tasks organized in this study are also provided. Here, “3D Healthy” refers to datasets containing healthy individuals with 3D MRI volumes, while “3D Tumor” refers to datasets containing brain tumor patients with 3D MRI volumes. The PubMed Central dataset includes both normal and brain tumor subjects. For ReMIND dataset, we utilized 108 brain tumor cases, excluding 6 brain disease cases.

Data Type	Data Resource	No. of subjects	No. of images	Original Information	Organized Tasks
3D Tumor	Brain metastase MRI dataset ¹²²	75	85,400	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	Brain-TR-GammaKnife ⁷³	47	8,300	Diagnosis	Tumor classification
3D Healthy	AOMIC-ID1000 ¹²³	928	1, 101, 652	Healthy	Tumor or healthy classification
3D Healthy	AOMIC-PIOP1 ¹²⁴	216	251, 380	Diagnosis	Tumor or healthy classification
3D Healthy	IXI ¹²⁵	582	398, 774	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (Listening Task) ¹²⁶	78	136, 694	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (Healthy Adults) ¹²⁷	66	122, 175	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (Visual Audiovisual Speech) ¹²⁸	60	105, 160	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (Dynamic Passive Threat) ¹²⁸	82	126, 299	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (UCLA Consortium) ¹²⁹	272	285, 910	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (NIMH Healthy) ¹³⁰	157	461, 281	Diagnosis	Tumor or healthy classification
3D Healthy	DLBS dataset ¹³¹	315	99, 344	Diagnosis	Tumor or healthy classification
3D Healthy	QTAB dataset ¹³²	422	1, 251, 467	Diagnosis	Tumor or healthy classification

Extended Data Table 3 | Characteristics of prospective dataset, including surgical group and non-operative group for each brain tumor type.

	Surgical Group				Non-operative group				Entire dataset
	No. of subjects	Sex		Age in years Mean $\pm$ Std (range)	No. of subjects	Sex		Age in years Mean $\pm$ Std (range)	
		Male	Female			Male	Female		
Total	639	290	359	48 $\pm$ 17 (2-83)	74	33	41	41 $\pm$ 21 (2-66)	713
MET	25	16	10	60 $\pm$ 8 (38-72)	7	4	3	45 $\pm$ 11 (34-55)	32
GCT	15	11	6	16 $\pm$ 9 (7-39)	3	2	1	8 $\pm$ 1 (7-9)	18
GGN	119	69	50	44 $\pm$ 19 (5-83)	18	8	10	30 $\pm$ 24 (2-60)	137
MEN	185	44	141	53 $\pm$ 11 (14-78)	22	6	16	50 $\pm$ 11 (29-66)	207
TSR	146	81	68	49 $\pm$ 16 (4-77)	17	9	8	54 $\pm$ 8 (37-63)	163
MNM	40	22	21	38 $\pm$ 21 (2-72)	2	1	1	15 $\pm$ 1 (14-16)	42
CPN	87	36	52	52 $\pm$ 13 (22-76)	5	3	2	54 $\pm$ 15 (42-62)	92
CPT	4	1	3	43 $\pm$ 15 (19-60)	-	-	-	-	4
HEM	10	5	5	64 $\pm$ 7 (54-74)	-	-	-	-	10
EMB	7	5	2	17 $\pm$ 12 (5-35)	-	-	-	-	7
PIN	1	0	1	43	-	-	-	-	1

In this table, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).

Extended Data Table 4 | The designed special tokens for data differentiation and task-driven prompt design to distinguish between different tasks. A 2D MRI slice is enclosed with '<image>' and '</image>' tags, while 3D MRI volumes use '<vid>' and '</vid>' tags. The and <vi> special tokens represent the 2D image features and 3D volume features, respectively. Additionally, a special token '<s>' is inserted between frames to aid in learning spatial dependencies within the 3D volume.

Tasks	Visual Tokens	Task-specific Instructions Example	Sample Output
2D MRI slice level tumor classification	<image></image>	Is a tumor present in this MRI slice? If so, what type?	<tumor_i>
2D MRI slice-text description	<image></image>	Please make a description for this image.	A lesion in left lobe, showing T1 hyperintense
3D MRI tumor classification	<vid1><v1><v1>...</vid1> <vid2><v2><v2>...</vid2> ...	For a 6-year-old male patient, is a tumor present in these MRIs? If so, what type?	<tumor_i>
3D MRI-report generation	<vid1><v1><v1>...<vid1><s> <vid2><v2><v2>...</vid2><s> ...	For a 6-year-old male patient, please generate a radiology report for him.	In the ..., hyperintense... midline shift to ...
3D MRI report generation tumor classification	<vid1><v1><v1>...</vid1><s> <vid2><v2><v2>...</vid2><s> ...	For a 6-year-old male patient, please generate a radiology report and make a diagnosis.	In the ..., hyperintense...; <tumor_i>
Uncertainty quantification	<vid1><v1><v1>...</vid1><s> <vid2><v2><v2>...</vid2><s> ...	For a 6-year-old male patient, please generate a radiology report, make a diagnosis, and generate reliable score for your diagnosis.	In the ..., hyperintense... midline shift to ... <tumor_i>, confidence: 80%
Molecular subgroup prediction	<vid1><v1><v1>...</vid1><s> <vid2><v2><v2>...</vid2><s> ...	For a 6-year-old male glioma patient, please make a diagnosis, classifying the glioma's subgroup.	This is IDH ... mutant type

Extended Data Table 5 | Diagnostic performance comparison between full MRI sequences (T1, T1c, T2, FLAIR) input and excluding one of the four standard sequences, evaluated on 3,571 patients from the primary and external test datasets. Only patients with complete MRI data across all four sequences were included to eliminate potential confounding due to missing sequences. The remaining 1,555 patients in the original primary and external test datasets with missing sequences were excluded from the analysis.

	F_1 score				
	Full Sequence	Excluded T1	Excluded T1c	Excluded T2	Excluded FLAIR
MET	0.58	0.52	0.45	0.54	0.53
GCT	0.64	0.48	0.47	0.58	0.53
GGN	0.83	0.80	0.72	0.82	0.82
MEN	0.92	0.89	0.82	0.89	0.89
TSR	0.75	0.65	0.52	0.63	0.63
MNM	0.56	0.43	0.44	0.45	0.48
CPN	0.81	0.76	0.64	0.79	0.78
CPT	0.44	0.27	0.27	0.43	0.27
HEM	0.58	0.43	0.42	0.42	0.45
EMB	0.67	0.44	0.4	0.42	0.44
PIN	0.43	0.36	0.22	0.43	0.29
MEL	0.17	0.17	0.17	0.17	0.17
Frequency-weighted F_1	0.82	0.78	0.71	0.79	0.79

In this table, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.

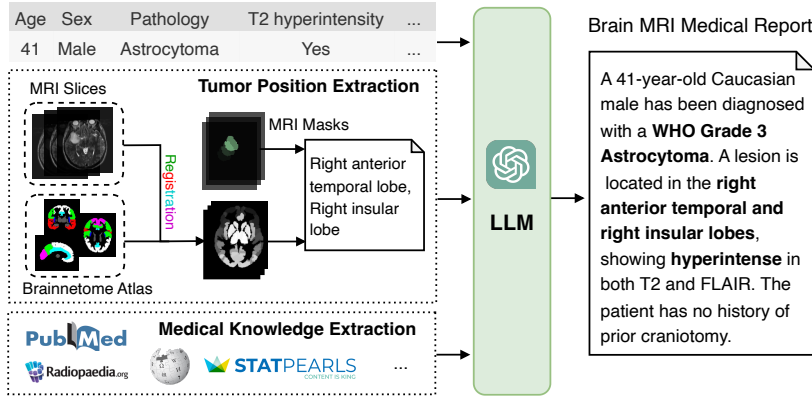
Extended Data Table 6 | Comparison of F_1 scores between our BrainVLM model, neuroradiologists, and three state-of-the-art AI models (RadFM³¹, Merlin³², and ChatGPT-4o³³). We evaluated 3,797 patients from the primary test dataset and 1,329 patients from the external test dataset, respectively.

	F_1 score (Primary Testing)					F_1 score (External Testing)				
	Ours	Doctor	RadFM ³¹	Merlin ³²	ChatGPT-4o ³³	Ours	Doctor	RadFM ⁹⁶	Merlin ³²	ChatGPT-4o ³³
MET	0.53	0.56	0.45	0.25	0.04	0.61	0.52	0.44	0.40	0.10
GCT	0.57	0.51	0.33	0.15	0.08	0.73	0.52	0.30	0.10	0.11
GGN	0.83	0.82	0.68	0.59	0.47	0.8	0.70	0.45	0.56	0.38
MEN	0.91	0.91	0.64	0.73	0.61	0.82	0.83	0.53	0.63	0.41
TSR	0.88	0.88	0.75	0.49	0.56	0.87	0.84	0.39	0.55	0.48
MNM	0.56	0.43	0.30	0.12	0.03	0.46	0.55	0.27	0.0	0.0
CPN	0.78	0.74	0.40	0.27	0.17	0.8	0.74	0.28	0.30	0.11
CPT	0.54	0.51	0.19	0.36	0.08	0.29	0.18	0.0	0.0	0.0
HEM	0.64	0.45	0.31	0.15	0.0	0.41	0.39	0.33	0.28	0.0
EMB	0.68	0.58	0.47	0.29	0.13	0.48	0.42	0.26	0.25	0.0
PIN	0.41	0.11	0.27	0.17	0.0	0.36	0.17	0.0	0.0	0.0
MEL	0.22	0.12	0.0	0.0	0.0	-	-	-	-	-
Weighted F_1	0.83	0.81	0.62	0.54	0.37	0.76	0.71	0.43	0.52	0.30

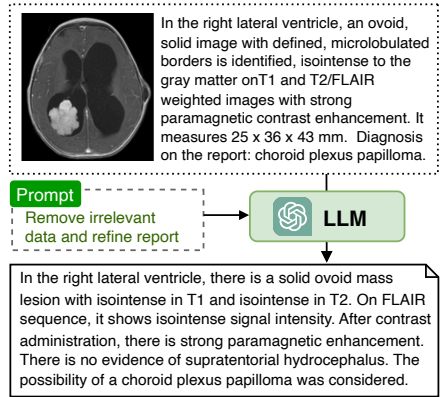
In this table, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.

a. Pipeline for Processing Structured Repositories and Pre-cured Datasets

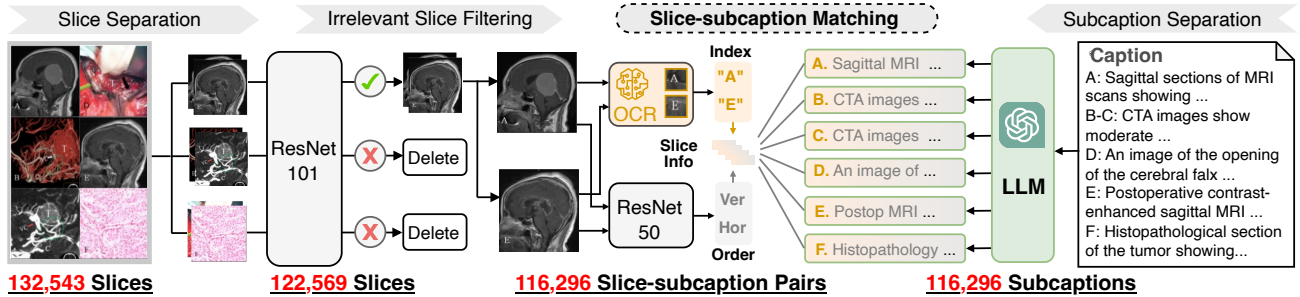
a1. Datasets with Tumor Masks



a2. Datasets with Raw Radiology Report



b. Pipeline for Processing Unstructured Web-based Resources

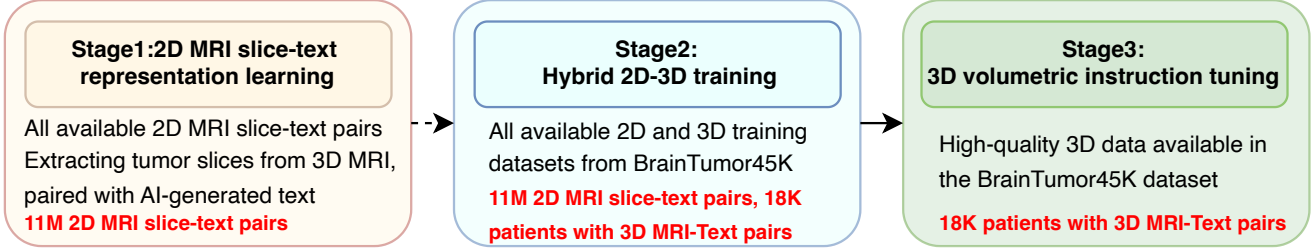


c. Pipeline for Processing Private Medical Institutions

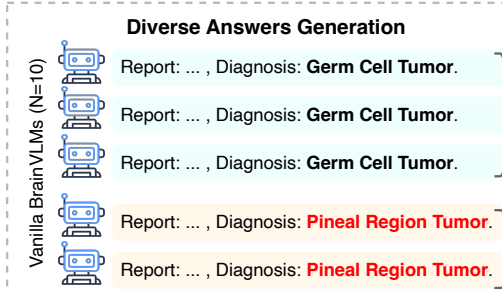


Extended Data Fig. 1 | Preprocessing pipeline for BrainTumor46K. a. The preprocessing pipeline for structured repositories and pre-cured datasets. **a1:** For pre-cured datasets containing patient metadata and tumor masks, multi-sequence MRIs and their corresponding segmentation masks were co-registered to a standard neuroanatomical atlas to localize tumors. All available metadata—including imaging modality, gender, diagnostic labels, and extracted tumor descriptions—were utilized to generate comprehensive medical reports using ChatGPT-4o, prompted with RAG techniques. **a2:** For datasets comprising multi-parametric MRI scans and accompanying radiology reports (Radiopaedia), irrelevant information was removed from the original reports using prompt-based filtering with ChatGPT-4o.¹³³ **b.** Preprocessing workflow for unstructured web-based resources, such as PubMed. Using PubMed Central as an example, we illustrate the preprocessing strategy for extracting MRI slice-subcaption pairs. First, brain tumor MRI figures were retrieved through keyword search, and irrelevant images were filtered using a ResNet-101 classifier. Second, compound images were decomposed into individual slices with a Faster R-CNN detector. Third, subcaptions were associated with image slices using Google OCR and ChatGPT-4o. The numbers of MRI slices and resulting slice-subcaption pairs are also summarized below. **c.** The preprocessing pipeline for institutional medical data. Chinese radiology reports were trimmed to retain modality-specific information, reformatted to the standard template, translated into English, and quality-checked by clinicians and multiple large-language models.

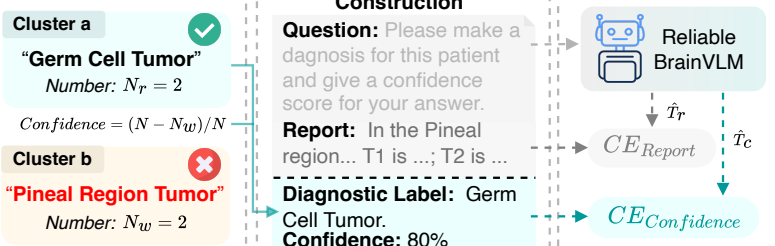
a. Three-stage progressive training strategy of BrainVLM



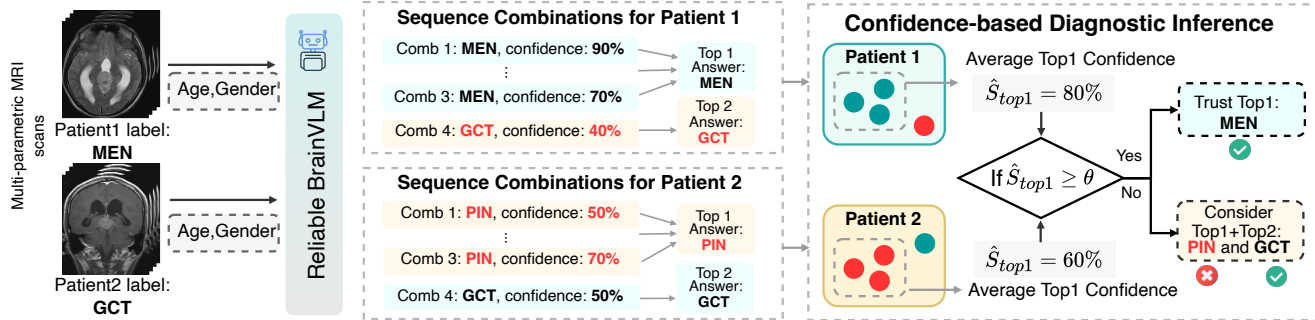
b. Consensus-driven Response Clustering



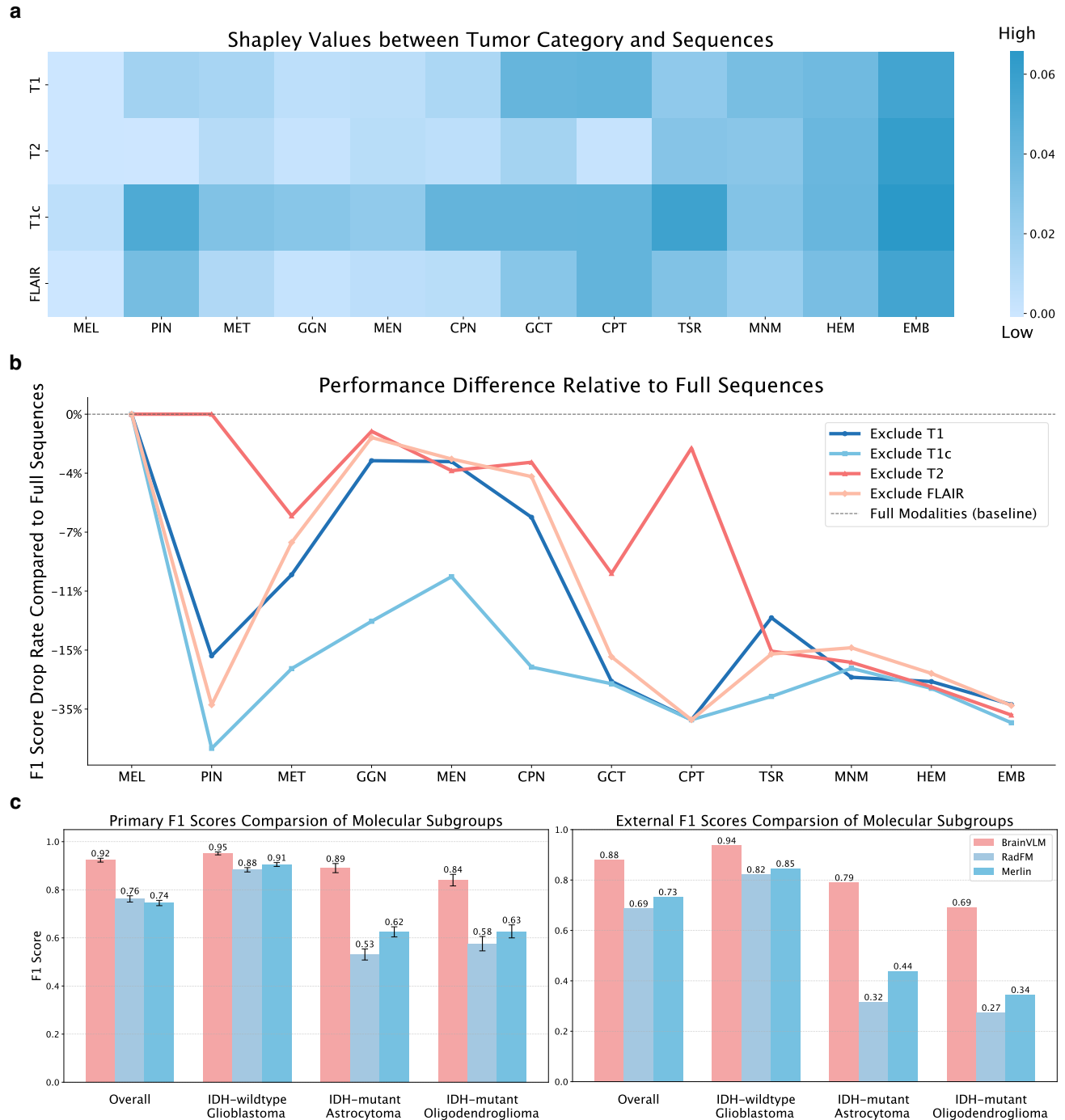
c. Diagnostic Reliability Training



d. Reliability-driven Inference of Top2 Diagnostic Candidates

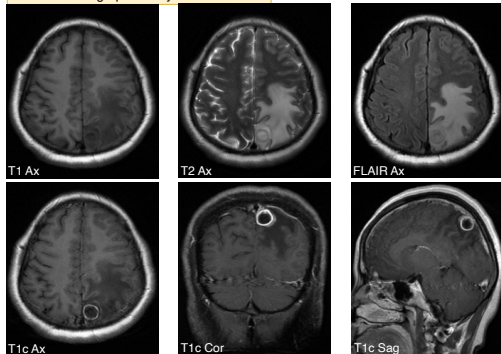


Extended Data Fig. 2 | Three-stage progressive training strategy and uncertainty quantification implementation of BrainVLM, including the confidence-triggered Top 2 supplementary diagnosis inference pipeline. **a. Three-stage progressive training strategy of BrainVLM.** BrainVLM was developed using a three-stage progressive training approach: (1) 2D MRI slice-text representation learning using all curated 2D MRI slice-text pairs; (2) volumetric context integration via hybrid 2D-3D training with all available 2D and 3D training datasets; and (3) comprehensive 3D volumetric instruction tuning leveraging the high-quality 3D data in BrainTumor46K. **b,c. Uncertainty quantification implementation pipeline.** The uncertainty quantification of our BrainVLM consists of two steps: consensus-driven reliability dataset construction and confidence-aware reliable model finetuning. **b. Consensus-driven reliability dataset construction.** A reliability dataset is constructed by evaluating N vanilla BrainVLM models on the training dataset and clustering their diagnostic answers to derive confidence scores. **c. Confidence-aware reliable BrainVLM finetuning.** The consensus-derived reliability knowledge in the reliability dataset is distilled into reliable BrainVLM through confidence-aware reliable fine-tuning. **d. Confidence-triggered Top 2 supplementary diagnosis inference.** During deployment, each prediction made by the reliable BrainVLM is accompanied by a confidence score. If the confidence score of the Top 1 prediction falls below a predefined threshold (set at 75% in this study), the Top 2 prediction is also reported as a supplementary diagnosis to enhance clinical robustness. In this figure, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).



Extended Data Fig. 3 | Impact of individual MRI sequences on BrainVLM diagnostic accuracy for brain tumor classification. Diagnostic performance was evaluated in 3,571 patients by comparing the full-sequence model (T1, T1c, T2, FLAIR) to models in which each sequence was individually omitted. Only patients with complete MRI data across all four sequences were included to eliminate potential confounding due to missing sequences. The remaining 1,555 patients in the original primary and external test datasets with missing sequences were excluded from the analysis. For each tumor type, the performance drop rate (measured in F_1 score) resulting from the exclusion of a specific modality is shown (b, central panel). Both the direct performance drop rate and Shapley value analysis (a, top panel) highlight distinct sequence dependencies among tumor classes: T1c removal resulted in the greatest performance decline across all tumor types, underscoring its essential role. EMB classification was uniformly sensitive to the absence of any sequence (35% mean drop), while GGN and MEN predictions remained stable when T2 or FLAIR or T1 was missing. In contrast, CPT predictions were more dependent on T1, T1c and FLAIR, but showed resilience to the exclusion of T2. c, Molecular subgroups F1 scores comparison between our BrainVLM with Merlin and RadFM. In this figure, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.

Patient Demographic: 50-year-old Female



Significant Omission

Model Prediction with Confidence Score and Radiology Report

Findings:

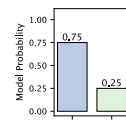
In left parietal lobe, there is a nodular lesion with **hypointense** (Missing of description of isointense) in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **significant enhancement**. Compression of the left lateral ventricle is observed. Midline structures shift to the right.

Impression:

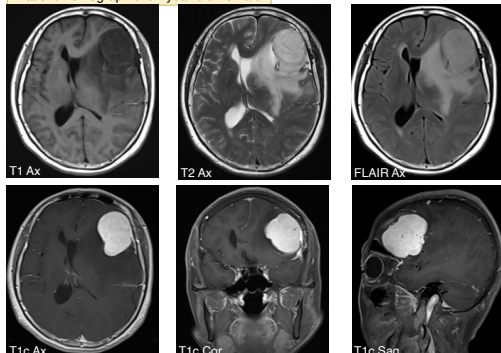
A nodular lesion was observed in **left parietal lobe**, consider the possibility of **brain metastase tumor** (Confidence 75%).

Gold Standard Pathology Labels

Brain Metastase tumor



Patient Demographic: 54-year-old Female



Model Prediction with Confidence Score and Radiology Report

Findings:

In left frontal lobe, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a (homogeneous) **significant enhancement**. Compression of the left lateral ventricle is observed. Midline structures shift to the right.

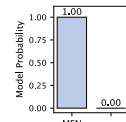
Impression:

A mass lesion was observed in the **left frontal lobe**, considering the possibility of **meningioma**. (Confidence 100%).

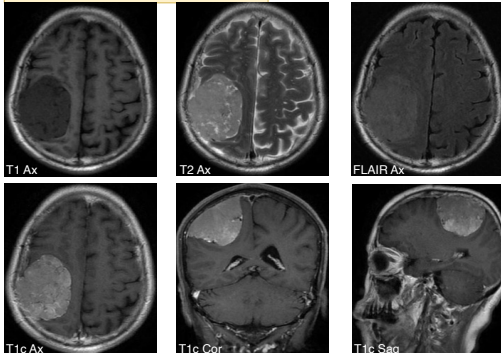
Gold Standard Pathology Labels

(Left frontal lobe)

Meningioma



Patient Demographic: 67-year-old female



Model Prediction with Confidence Score and Radiology Report

Finding

In right parietal lobe, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **significant enhancement**. Compression of the right lateral ventricle is observed. Midline structures shift to the left. This patient was diagnosed with Meningioma.

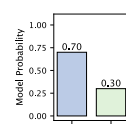
Impression:

A mass lesion was observed in the **right parietal lobe**. Consider the possibility of **meningioma** (Confidence 70%) and brain metastase tumor (Confidence 30%).

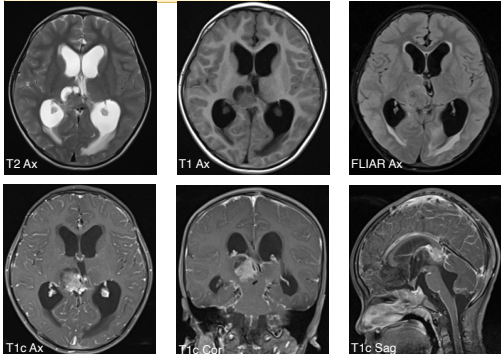
Gold Standard Pathology Label

(Right parietal lobe)

Meningioma



Patient Demographic: 6-year-old female



Model Prediction with Confidence Score and Radiology Report

Findings:

In pineal region, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **significant enhancement of the solid component**. Dilation of bilateral ventricles is observed. (Compression of the third ventricle is observed). No midline structure shift.

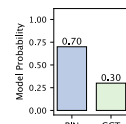
Impression

A mass lesion was observed in **pineal region**, considering the possibility of **pineal region tumor** (Confidence 70%) and germ cell tumor (Confidence 30%).

Gold Standard Pathology Label

(Pineal region)

Pineal region tumor



Significant Error

Radiologist Report

Finding: In left parietal lobe, there is a mass lesion with hypointense-isointense in T1, hyperintense signal in T2 and FLAIR. After contrast administration, there is a significant ring-like enhancement. Compression of the left lateral ventricle is observed. Midline structures shift to the right.
Impression: A mass lesion in the parietal lobe, consider **brain metastase tumor**.

GPT-4o Generated Report

Finding: In the **left frontal lobe**, there is a well-circumscribed mass lesion demonstrating **hypointensity** on T1-weighted images and **hyperintensity** on T2-weighted/FLAIR sequences. After contrast administration, there is uniform **enhancement**. Midline shift to the **left**.
Impression: A mass lesion in the **left frontal lobe**, consider **meningioma**.

Radfm Generated Report

Finding: In **left parietal lobe**, there is a mass lesion with **hypointense-isointense** in T1, **heterogenous** signal in T2, **hyperintense** in FLAIR. After contrast administration, there is **significant enhancement**. Midline structures shift to the **right**. (Missing of compression description)
Impression: A mass lesion was observed in **left parietal lobe**, considering **meningioma**.

Radiologist Report

Finding: In left frontal lobe, there is a mass lesion with hypointense in T1, hyperintense in T2, hyperintense in FLAIR. After contrast administration, there is a homogeneous enhancement. Compression of the lateral ventricles is observed. Midline structures shift to the right.
Impression: A mass lesion in the left frontal lobe, consider **mesenchymal, non-meningothelial tumour**.

GPT-4o Generated Report

Finding: In the **right frontal region**, there is a well-circumscribed extra-axial mass lesion demonstrating **hypointensity** in T1 and **hyperintensity** in T2, with **homogeneous contrast enhancement**. Surrounding vasogenic edema is evident as **hyperintensity** on FLAIR images. Midline shift is **not noted**. Ventricular obstruction is **not noted**.
Impression: A mass lesion in **right frontal region**, consider **meningioma**.

Radfm Generated Report

Finding: In **Right frontal lobe**, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a (homogeneous) **marked enhancement**. **No significant dilatation** of the ventricular system. **No midline structure shift**.
Impression: A mass lesion in **right frontal lobe**, consider **brain metastase tumor**.

Radiologist Report

Finding: In right parietal lobe, there is a mass lesion with hypointense in T1, hyperintense in T2, hyperintense in FLAIR. After contrast administration, there is a marked enhancement. Compression of the right lateral ventricle is observed. Midline structures shift to the left.
Impression: A mass lesion in the right parietal lobe, consider **meningioma**.

GPT-4o Generated Report

Finding: In **left temporal lobe**, there is a mass lesion with **hypointense** signal in T1, **hyperintense** signal in T2, **hyperintense** signal in FLAIR. After contrast administration, there is **heterogeneous** significant enhancement. Compression of the **right lateral ventricle** is observed. **No midline shifts**.
Impression: A mass lesion found in **left temporal lobe**, consider **glioma**.

Radfm Generated Report

Finding: In right frontal lobe, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **marked enhancement**. **No significant dilatation** of the ventricular system. **No midline structure shift**.
Impression: A mass lesion was observed in **right frontal lobe**, consider **brain metastase tumor**.

Radiologist Report

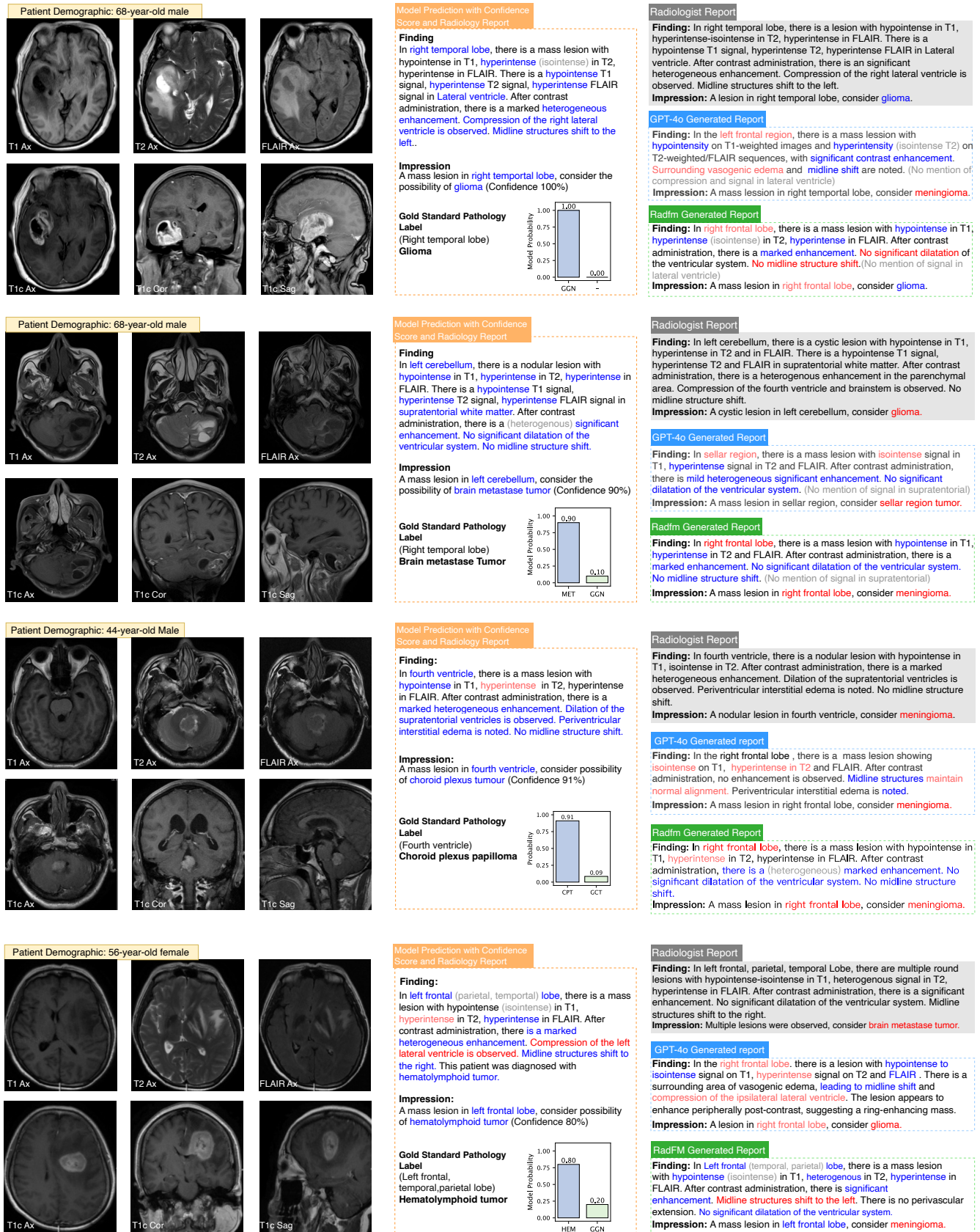
Finding: In right pineal region, there is a lesion with hypointense in T1, hyperintense in T2, hyperintense in FLAIR. After contrast administration, there is a significant enhancement of the solid component. Dilation of bilateral ventricles is observed. Compression of the third ventricle is observed.
Impression: A lesion located in right pineal region, consider **germ cell tumor**.

GPT-4o Generated Report

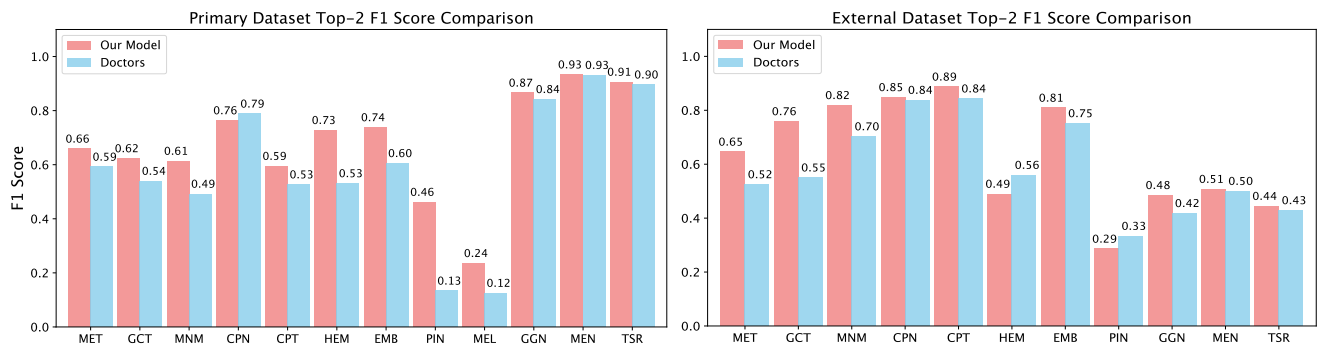
Finding: In the **occipital horns of the lateral ventricles**, there is a mass lesion with **hypointense** signal in T1, **hyperintense** signal in T2 and FLAIR. After contrast administration, there is **significant enhancement**. No midline shifts.
Impression: A mass lesion in the **lateral ventricles**, consider **glioma**.

Radfm Generated Report

Finding: In **right frontal lobe**, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **marked enhancement**. **No significant dilatation** of the ventricular system. **No midline structure shift**.
Impression: A mass lesion in the **right frontal lobe**, consider **germ cell tumor**.



Extended Data Fig. 4 | Examples of reports and diagnoses generated by our BrainVLM, compared with those from doctors (expert-curated radiology report) and other state-of-the-art AI models (RadFM³¹ and ChatGPT-4o³³). For each diagnosis generated by BrainVLM, a corresponding reliability score is also provided. We use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematomphoid tumors.



Extended Data Fig. 5 | F_1 score comparison across 12 tumor types, evaluating the Top 2 predictions of our model and neuroradiologists on both primary and external datasets. In this figure, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).